PHYS 357: Honours Quantum Physics 1

W.R.YCH.

August 31st, 2026

Abstract

Contents

1 Introduction **1**

2 Mathematical Background **1**
2.1 Taylor series . . . 1
2.2 Euler's formula and trigonometric identities . . . 2
2.3 Bases . . . 2

3 Stern-Gerlach Experiments **2**
3.1 The classical picture . . . 2
3.1.1 Angular momentum . . . 2
3.1.2 Magnetic moment of a circulating charge . . . 3
3.1.3 Gyromagnetic ratio . . . 3
3.2 Spin . . . 3
3.2.1 Intrinsic moment and the g-factor . . . 3
3.2.2 Magnetons . . . 4
3.3 The Stern-Gerlach apparatus . . . 5
3.3.1 Force in a nonuniform field . . . 5
3.3.2 Silver atoms . . . 5
3.3.3 Classical prediction versus observation . . . 6
3.4 Spin quantization . . . 6
3.5 Sequential Stern-Gerlach experiments . . . 6
3.5.1 Experiment 1: SGz then SGz . . . 7
3.5.2 Experiment 2: SGz then SGx . . . 7
3.5.3 Experiment 3: SGz, SGx, then SGz . . . 7
3.5.4 Experiment 4: SGz, modified SGx, then SGz . . . 8

4 Quantum States **8**
4.1 Kets, bras and bases . . . 8
4.2 Measurement and probability . . . 9
4.2.1 Bras of superpositions . . . 10
4.2.2 Normalization . . . 10
4.2.3 Global and relative phase . . . 10
4.3 Expectation value and uncertainty . . . 11
4.3.1 Measured averages and shot noise . . . 11
4.4 Spin states along x and y . . . 11
4.4.1 The state $|+\hat{x}\rangle$ . . . 11
4.4.2 Experiment 5: measuring S_y . . . 12
4.4.3 Fixing the phases . . . 12

5 Matrix Mechanics **13**
5.1 States as column vectors . . . 13
5.2 Operators as matrices . . . 13
5.3 Observables as operators . . . 14
5.3.1 Spin operators and Pauli matrices . . . 15

5.3.2 Hermitian operators . . . 16
5.4 Change of basis . . . 16
5.4.1 Operators in another basis . . . 18

6 Rotations **18**
6.1 The rotation operator . . . 18
6.2 Generators of rotations . . . 19
6.3 Functions of an operator . . . 19
6.3.1 Rotations about $\hat{z}$ . . . 20
6.4 Rotations about any axis . . . 20

7 Angular Momentum **21**
7.1 Non-commuting rotations and commutation relations . . . 21
7.2 Compatible observables . . . 22
7.3 Uncertainty relations . . . 23
7.4 Total angular momentum . . . 24
7.5 Ladder operators . . . 25
7.6 Matrix elements . . . 26
7.7 Rotations by 2π: fermions and bosons . . . 26
7.8 Atomic quantum numbers . . . 27

8 Time Evolution **27**
8.1 The time-evolution operator . . . 27
8.2 Stationary states . . . 28
8.3 Time dependence of expectation values . . . 29
8.4 Spin precession . . . 29
8.4.1 Expectation values . . . 30
8.4.2 From the equation of motion . . . 31
8.5 Magnetic resonance . . . 31
8.5.1 The rotating frame . . . 32
8.5.2 Rotating-wave approximation . . . 33
8.5.3 Rabi oscillations . . . 33

A Solutions **35**

B Useful Links **37**

1 Introduction

Textbook used is JS Townsend, *A Modern Approach to Quantum Mechanics*, 2nd ed., University Science Books 2012. Course grading is divided as follows:

- 10% problem sets (must be handwritten scans).
- 40% quizzes (on Wednesdays, best 8 of 12).
- 20% midterms (best of 2, October 2nd and November 6th).
- 30% final exam.

2 Mathematical Background

2.1 Taylor series

Definition 1 (Taylor series)**.** *A function f that is smooth near a (it can be differentiated any number of times there) equals the power series*

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n, \tag{1}$$

where $f^{(n)}$ is the nth derivative. The special case $a = 0$ is called the Maclaurin series.

Result 1 (Series used in this course)**.**

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}, \qquad \cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}, \qquad \sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}, \tag{2}$$

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n \quad (|x|<1), \qquad (1+u)^N = \sum_{n=0}^{N} \binom{N}{n} u^n \quad (N \in \mathbb{N}), \tag{3}$$

$$\ln(1+u) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} u^n}{n} = u - \frac{u^2}{2} + \frac{u^3}{3} - \cdots \quad (|u|<1). \tag{4}$$

Problem 1 (The exponential as a limit)**.** *Show that* $\lim_{N\to\infty} (1 + x/N)^N = e^x$.

Remark 1. *This limit is what lets a rotation be written as an exponential (Section 6.2).*

2.2 Euler's formula and trigonometric identities

Result 2 (Euler's formula)**.** *Comparing the series for $e^{i\theta}$, $\cos\theta$ and $\sin\theta$ gives $e^{i\theta} = \cos\theta + i\sin\theta$, hence*

$$\cos\theta = \frac{e^{i\theta} + e^{-i\theta}}{2}, \qquad \sin\theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}. \tag{5}$$

These are used constantly when rotation matrices are written out.

Result 3 (Double-angle identities)**.** *Squaring $e^{ix} = \cos x + i\sin x$ and comparing with $e^{2ix} = \cos 2x + i\sin 2x$,*

$$\cos^2 x - \sin^2 x = \cos 2x, \qquad 2\sin x\cos x = \sin 2x. \tag{6}$$

These turn the half angles $\phi/2$ in spin rotation matrices back into full angles ϕ.

2.3 Bases

Definition 2 (Basis)**.** *A set of vectors is* linearly independent *if none of them can be written as a combination of the others, and it* spans *a space if every vector in the space is some combination of them. A* basis *is a set that does both, so every vector is a unique combination of the basis vectors. The number of basis vectors is the dimension of the space.*

Definition 3 (Kronecker delta)**.**

$$\delta_{ij} = \begin{cases} 1 & i = j, \\ 0 & i \neq j. \end{cases} \tag{7}$$

3 Stern-Gerlach Experiments

August 31 - September 9, 2026

3.1 The classical picture

3.1.1 Angular momentum

Definition 4 (Orbital angular momentum)**.** *The orbital angular momentum of a particle is*

$$\vec{L} = \vec{r} \times \vec{p}, \tag{8}$$

where $\vec{r}$ is the position vector and $\vec{p} = m\vec{v}$ is the linear momentum.

For motion confined to the x-y plane, only the z component survives. On a circular path of radius r with speed v,

$$L_z = mvr.$$

Definition 5 (Angular momentum of a rigid body)**.** *For a rigid body*[1] *rotating about a fixed axis,*

$$L = I_0\omega,$$

with ω the angular velocity and I_0 the moment of inertia, which measures how the mass is distributed relative to the rotation axis.

[1] A rigid body is one that does not deform under the action of external forces.

3.1.2 Magnetic moment of a circulating charge

Definition 6 (Magnetic moment)**.** *A vector $\vec{\mu}$ measuring the net circulation of electric charge. It sets the torque, and in a nonuniform field the force, that a magnet experiences in an external magnetic field.*

A point particle of mass m and charge q on a circular orbit of radius r and period T passes any given point once per period, so it acts as a steady current $I = q/T = qv/2\pi r$. Its moment is

$$\mu = \frac{IA}{c}^{2} = \frac{1}{c}\left(\frac{qv}{2\pi r}\right)\left(\pi r^2\right) = \frac{q}{2mc}\, mvr = \frac{q}{2mc}\, L_z,$$

where A is the area enclosed by the orbit.

3.1.3 Gyromagnetic ratio

Definition 7 (Gyromagnetic ratio)**.** *The constant γ relating magnetic moment to angular momentum,*

$$\vec{\mu} = \gamma \vec{L}, \qquad \gamma = \frac{q}{2mc} \text{ for a circulating charge.} \tag{9}$$

Remark 2. *This is purely classical: a point charge on a definite orbit, nothing quantized. It matters because measuring $\vec{\mu}$ is how we measure angular momentum, which is what the Stern-Gerlach apparatus does.*

3.2 Spin

Definition 8 (Spin)**.** *Spin $\vec{S}$ is an* intrinsic *angular momentum: a fixed property of the particle type, like charge or mass, not caused by any motion through space. A stationary electron alone in empty space still has spin. It comes in addition to any orbital $\vec{L}$, not in place of it.*

3.2.1 Intrinsic moment and the g-factor

Spin also carries a magnetic moment proportional to it, but with an extra dimensionless factor,

$$\vec{\mu} = \gamma \vec{S}, \qquad \gamma = \frac{gq}{2mc}. \tag{10}$$

Definition 9 (g-factor)**.** *The dimensionless number g above. It says how much stronger or weaker the moment is than the classical value ($g = 1$). It must be measured or derived from relativistic quantum theory.*

[2] Gaussian (CGS) units, where a current loop has moment IA/c. In SI units it is IA.

3.2.2 Magnetons

Definition 10 (Planck constant)**.** *The constant that sets the scale at which quantization matters. It relates photon energy to frequency, $E = h\nu$, and*

$$h \approx 6.626 \times 10^{-34}\ \mathrm{J\,s}, \qquad \hbar = \frac{h}{2\pi} = 1.055 \times 10^{-27}\ \mathrm{erg\,s} = 6.582 \times 10^{-16}\ \mathrm{eV\,s}.$$

Both have units of angular momentum, $[\hbar] = [L_z] = \mathrm{kg\,m^2/s}$.

Writing $\vec{S} = \hbar(\vec{S}/\hbar)$ with $\vec{S}/\hbar$ dimensionless, the $\hbar$ joins the gyromagnetic ratio. The combination $|e|\hbar/2mc$ then carries all the units and sets the natural size of the moment for a given mass.

Definition 11 (Magnetons)**.** *The natural units of magnetic moment for the electron (Bohr magneton) and for nucleons, i.e. protons and neutrons (nuclear magneton):*

$$\mu_B = \frac{|e|\hbar}{2m_e c}, \qquad \mu_N = \frac{|e|\hbar}{2m_p c} \approx \frac{\mu_B}{1836}, \tag{11}$$

since $m_p/m_e \approx 1836$.

The magnetic moments are then

$$\vec{\mu}_e = -g_e\,\mu_B\,\frac{\vec{S}}{\hbar}, \qquad \vec{\mu}_{p,n} = g_{p,n}\,\mu_N\,\frac{\vec{S}}{\hbar}, \tag{12}$$

where the minus sign for the electron comes from its negative charge.

Particle	Charge q	Mass m	g-factor	Gyromagnetic ratio γ
Electron	$-\lvert e\rvert$	m_e	$g_e \approx 2$	$-g_e\mu_B/\hbar$
Proton	$+\lvert e\rvert$	m_p	$g_p \approx 5.59$	$g_p\mu_N/\hbar$
Neutron	0	$m_n \approx m_p$	$g_n \approx -3.83$	$g_n\mu_N/\hbar$

Table 1: Charges, masses, g-factors and gyromagnetic ratios.

Remark 3. *Classically $\mu = (q/2mc)L_z$ vanishes when $q = 0$, yet the neutral neutron has a moment. The reason is that the neutron is not a point particle: it is made of three charged quarks (one up quark with charge $+\frac{2}{3}|e|$ and two down quarks with $-\frac{1}{3}|e|$ each). The charges add to zero, but the quarks' spins and motion inside the neutron each give a moment, and these moments do not cancel. So its $\vec{\mu}$ cannot come from a single circulating point charge: a first sign the classical picture fails. The same internal structure is why $g_p \approx 5.59$ is far from 2.*

Remark 4. *Since $m_p \gg m_e$, $\mu_N \ll \mu_B$, so nuclei are deflected far less than electrons in a Stern-Gerlach device.*

3.3 The Stern-Gerlach apparatus

Definition 12 (Stern-Gerlach (SG) apparatus)**.** *A device that measures one component of a particle's magnetic moment. A beam of neutral atoms passes through a magnet shaped so that the field strength changes along a chosen axis, here z (a field* gradient*). The gradient pushes each atom by an amount proportional to μ_z, and a screen records where it lands. Since $\vec{\mu} = \gamma \vec{S}$, the landing position measures S_z.*

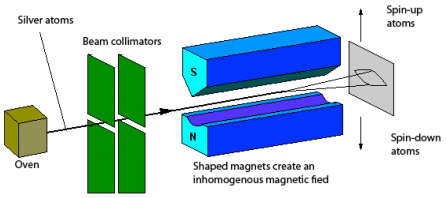


Figure 1: Illustration of the Stern-Gerlach apparatus.

3.3.1 Force in a nonuniform field

A magnetic moment in a field has potential energy $U = -\vec{\mu} \cdot \vec{B}$, so

$$\vec{F} = -\vec{\nabla} U = \vec{\nabla}(\vec{\mu} \cdot \vec{B}), \qquad F_z = \mu_z \frac{\partial B_z}{\partial z}.$$

Remark 5. *A uniform field gives no net force, only a torque (a twist that turns $\vec{\mu}$ without moving the atom). That is why the magnet needs a gradient.*

Definition 13 (Deflection)**.** *The sideways displacement of an atom on the screen caused by F_z. It is proportional to μ_z, so the screen pattern is a record of the moments in the beam.*

3.3.2 Silver atoms

Silver has zero net orbital angular momentum, $\vec{L} = 0$, and its whole moment comes from one unpaired outermost (valence) electron, so the atom behaves like a single electron spin. With

$$\mu_z = -\frac{g_e |e|}{2 m_e c} S_z$$

and a gradient $\partial B_z / \partial z < 0$,

$$F_z = g_e \frac{|e|}{2 m_e c} \left| \frac{\partial B_z}{\partial z} \right| S_z.$$

Result 4. *$F_z \propto S_z$: the deflection measures the z component of spin directly.*

3.3.3 Classical prediction versus observation

Atoms leave the oven with spins pointing in random directions, so $S_z = |\vec{S}|\cos\theta$ takes every value in $[-|\vec{S}|, |\vec{S}|]$. Classically the screen should show a continuous smear. It shows two spots.

Result 5 (Quantization of spin)**.** *The electron's spin component is quantized:*

$$S_z = \pm\frac{\hbar}{2}. \tag{13}$$

3.4 Spin quantization

Definition 14 (Spin quantum number)**.** *The number s, fixed by the particle type, taking values $s = 0, \frac{1}{2}, 1, \frac{3}{2}, \ldots$*[3] *A measurement of S_z gives*

$$S_z = m\hbar, \qquad m = -s, -s+1, \ldots, +s, \tag{14}$$

so the number of possible outcomes, and therefore of beams on the screen, is

$$n = 2s + 1. \tag{15}$$

Counting the beams gives $s = (n-1)/2$. Silver gives $n = 2$ beams, $S_z = \pm\hbar/2$, so the electron has $s = \frac{1}{2}$. The deuteron (a proton and a neutron with aligned spins) gives $n = 3$ beams, $S_z = +\hbar, 0, -\hbar$, so $s = 1$.

Remark 6. *S_z is a component. The magnitude is $|\vec{S}| = \hbar\sqrt{s(s+1)}$, which for the electron ($s = 1/2$) is $\hbar\sqrt{3}/2$.*

3.5 Sequential Stern-Gerlach experiments

We write SGz for a device whose gradient points along z, and SGx for the same device rotated by 90° about the beam axis.

Definition 15 (Ket)**.** *A ket $|\cdot\rangle$ is a label for a quantum state, treated as a vector. For electron spin, the two outcomes of an SGz give*

$$|+\hat{z}\rangle \equiv |\uparrow\rangle \text{ for } S_z = +\frac{\hbar}{2}, \qquad |-\hat{z}\rangle \equiv |\downarrow\rangle \text{ for } S_z = -\frac{\hbar}{2},$$

and likewise $|\pm\hat{x}\rangle$ for $S_x = \pm\hbar/2$, and so on for any axis. We follow Townsend and use the $|\pm\hat{n}\rangle$ labels.

Definition 16 (Filter)**.** *Each device sends atoms into two beams. A filter blocks one of them.*

Remark 7. *A beam straight from the oven always splits 50/50 in any SG device, whatever its orientation.*

[3] They follow from $[J_x, J_y] = i\hbar J_z$ via the ladder operators (Section 7.5), which force m to step by 1 from $-s$ to $+s$, so $2s$ is a whole number.

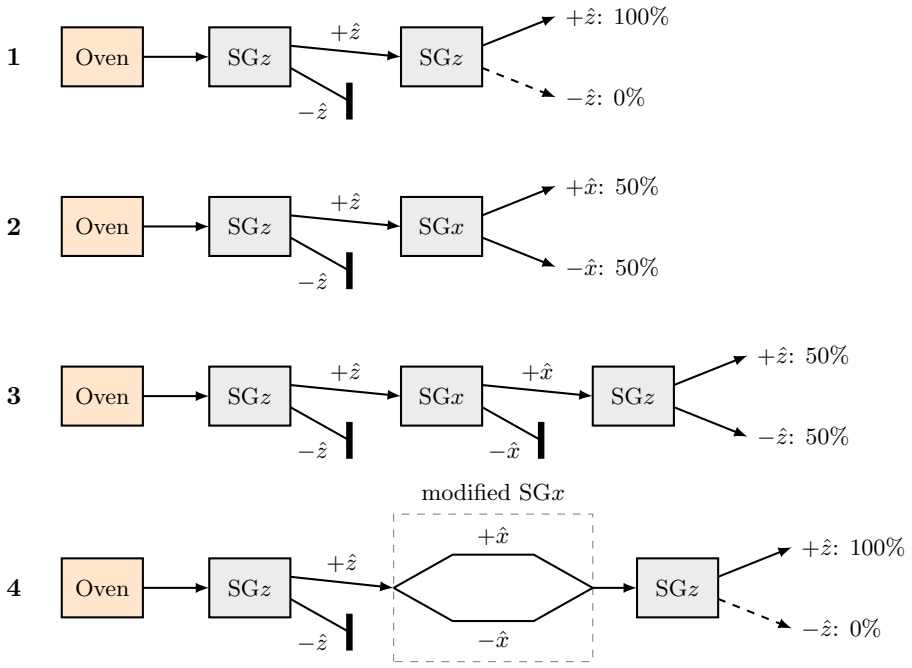


Figure 2: Sequential Stern-Gerlach experiments. Thick bars block a beam. Percentages are the fractions of atoms entering the last device. Every oven starts with N_0 atoms.

Remark 8. *The lecture notes use N_0 for the number of atoms entering the* second *device rather than leaving the oven, so their counts are twice the ones below.*

3.5.1 Experiment 1: SGz then SGz

Filter the $+\hat{z}$ beam and send it into a second SGz. Every atom comes out $+\hat{z}$. The measurement is repeatable: an atom found with $S_z = +\hbar/2$ gives the same answer again, so it left the first device in $|+\hat{z}\rangle$. We end with $N_0/2$ atoms in $|+\hat{z}\rangle$.

3.5.2 Experiment 2: SGz then SGx

Filter $+\hat{z}$ and send it into an SGx. Half come out $+\hat{x}$, half $-\hat{x}$: knowing S_z tells us nothing about S_x. We end with $N_0/4$ in each of $|\pm\hat{x}\rangle$.

3.5.3 Experiment 3: SGz, SGx, then SGz

Filter $+\hat{z}$, filter $+\hat{x}$, then measure S_z again. The beam splits 50/50 even though every atom was $+\hat{z}$ at the start. Blocking $+\hat{x}$ instead gives the same

result. We end with $N_0/8$ in each of $|\pm\hat{z}\rangle$.

Result 6. *No state has definite values of both S_x and S_z. Measuring S_x destroys what we knew about S_z.*

3.5.4 Experiment 4: SGz, modified SGx, then SGz

Definition 17 (Modified SG device)**.** *An SG device that splits the beam according to S_n and then recombines the two beams, so both outcomes exit at the same place without being recorded.*

Filter $+\hat{z}$, pass it through a modified SGx with neither path blocked, then measure S_z. Every atom comes out $+\hat{z}$: all $N_0/2$ end in $|+\hat{z}\rangle$.

Result 7. *The atoms cannot be going half through $+\hat{x}$ and half through $-\hat{x}$, since Experiment 3 showed each half alone gives 50/50. Instead the two paths* interfere*: as with water or light waves, contributions from the two paths add and can cancel, here cancelling the $-\hat{z}$ outcome. No definite path can be assigned to an atom. Section 4 makes this precise.*

Remark 9. *With one path blocked, a modified SG acts exactly like an ordinary SG.*

4 Quantum States

September 11-18, 2026

Throughout, an *n-level system* is one whose measurements have n possible outcomes. For spin s, $n = 2s + 1$, so spin-$\frac{1}{2}$ has $n = 2$.

4.1 Kets, bras and bases

Definition 18 (Bra and inner product)**.** *Every ket (Def. 15) $|\psi\rangle$ has a matching bra $\langle\psi|$, its conjugate transpose (adjoint): the ket's coefficients are complex conjugated, and the column becomes a row (Section 4.2.1). A bra followed by a ket gives a complex number, the inner product $\langle\phi|\psi\rangle \in \mathbb{C}$.*

Definition 19 (Orthonormal basis)**.** *The outcome states $\{|j\rangle : j = 1, \ldots, n\}$ of one measurement form a basis that is orthonormal:*

$$\langle i|j\rangle = \delta_{ij}. \tag{16}$$

For spin-$\frac{1}{2}$ this is $\langle\pm\hat{z}|\pm\hat{z}\rangle = 1$ and $\langle\pm\hat{z}|\mp\hat{z}\rangle = 0$, and the same holds for $\{|\pm\hat{x}\rangle\}$ or any other axis.

Remark 10. *The label j counts* outcomes*, not axes. Choosing to measure S_x gives $|1\rangle = |+\hat{x}\rangle$, $|2\rangle = |-\hat{x}\rangle$; choosing S_z gives $|1\rangle = |+\hat{z}\rangle$, $|2\rangle = |-\hat{z}\rangle$. Either way $n = 2$. A ket may also carry several labels, usually the quantum numbers that pin it down: $|s, m\rangle$ for spin, so $|\pm\hat{z}\rangle = |\frac{1}{2}, \pm\frac{1}{2}\rangle$, and for the deuteron $|1, +1\rangle, |1, 0\rangle, |1, -1\rangle$. Electrons in atoms are labelled the same way (Section 7.8).*

Definition 20 (Superposition)**.** *Any state is a linear combination of basis kets,*

$$|\psi\rangle = \sum_{j=1}^{n} c_j \, |j\rangle \,, \qquad c_j \in \mathbb{C}; \qquad \textit{spin-}\tfrac{1}{2}: \; |\psi\rangle = c_+ \, |+\hat{z}\rangle + c_- \, |-\hat{z}\rangle \,. \tag{17}$$

Proposition 1 (Coefficients are inner products)**.**

$$c_j = \langle j|\psi\rangle \,. \tag{18}$$

Proof. $\langle i|\psi\rangle = \sum_j c_j \langle i|j\rangle = \sum_j c_j \delta_{ij} = c_i$. □

Definition 21 (Projector)**.** *The outer product (a ket times a bra)* $P_j = |j\rangle \langle j|$*. It is an operator that keeps only the* $|j\rangle$ *part of a state,*

$$P_j \, |\psi\rangle = |j\rangle \langle j|\psi\rangle = c_j \, |j\rangle \,,$$

and projecting twice changes nothing: $P_j^2 = |j\rangle \langle j|j\rangle \langle j| = P_j$*.*

Proposition 2 (Completeness relation)**.** *Because every state can be expanded in the basis, the projectors sum to the identity:*

$$\mathbb{1} = \sum_{j=1}^{n} |j\rangle \langle j| \,; \qquad \textit{spin-}\tfrac{1}{2}: \; \mathbb{1} = |+\hat{n}\rangle \langle +\hat{n}| + |-\hat{n}\rangle \langle -\hat{n}| \;\; \textit{for any axis } \hat{n}. \tag{19}$$

Remark 11 (Inserting the identity)**.** *Writing* $\mathbb{1}$ *in this form in front of a state reproduces its expansion,* $|\psi\rangle = \sum_j |j\rangle \langle j|\psi\rangle$*. This trick is used in almost every derivation below.*

4.2 Measurement and probability

Definition 22 (Observable)**.** *A measurable quantity A. Its possible outcomes* a_j *each come with a state* $|a_j\rangle$*, and these states form an orthonormal basis. For spin-*$\frac{1}{2}$*, A can be* S_x*,* S_y *or* S_z*, with outcomes* $\pm\hbar/2$*.*

Principle 1 (Born rule)**.** *The probability that a system in* $|\psi\rangle$ *is found in* $|\phi\rangle$ *is*

$$P(\phi \,|\, \psi) = |\langle \phi|\psi\rangle|^2 \in [0, 1]. \tag{20}$$

In particular $P(A = a_j) = |\, \langle a_j|\psi\rangle \,|^2$*, e.g.* $P(S_z = \pm\hbar/2) = |c_\pm|^2$*.*

Definition 23 (Probability amplitude)**.** *The complex number* $\langle \phi|\psi\rangle$ *whose modulus squared is the probability.*

Remark 12. *The Born rule has no deeper derivation: it is a postulate that agrees with every experiment. As David Mermin put it, "shut up and calculate."*

4.2.1 Bras of superpositions

Write the bra with unknown coefficients, $\langle\psi| = c'_+ \langle+\hat{z}| + c'_- \langle-\hat{z}|$. By orthonormality,

$$\langle\psi|\psi\rangle = c'_+ c_+ + c'_- c_-.$$

A system in $|\psi\rangle$ is found in $|\psi\rangle$ with certainty, and it is always found in one of the two beams, so this must equal $|c_+|^2 + |c_-|^2 = 1$. That forces $c'_\pm = c^*_\pm$.

Result 8. *Going from a ket to its bra, conjugate the coefficients:*

$$\langle\psi| = \sum_j c_j^* \langle j|, \qquad \langle\phi|\psi\rangle = \langle\psi|\phi\rangle^*. \tag{21}$$

4.2.2 Normalization

Proposition 3 (Normalization)**.** $\langle\psi|\psi\rangle = 1$ *is the statement that the probabilities of all outcomes add to 1.*

Proof.

$$\langle\psi|\psi\rangle = \sum_{i,j} c_i^* c_j \langle a_i|a_j\rangle = \sum_j |c_j|^2 = \sum_j P(A = a_j) = 1.$$

□

Example. *For* $|\psi\rangle = \frac{1}{\sqrt{3}}|+\hat{z}\rangle + i\sqrt{\frac{2}{3}}|-\hat{z}\rangle$, $c_+ = 1/\sqrt{3}$ *and* $c_- = i\sqrt{2/3}$, *so* $P(S_z = +\hbar/2) = 1/3$ *and* $P(S_z = -\hbar/2) = |i|^2 \cdot 2/3 = 2/3$. *They add to 1, so the state is normalized.*

4.2.3 Global and relative phase

Definition 24 (Global and relative phase)**.** *A* global *phase multiplies the whole ket:* $e^{i\alpha}|\psi\rangle$ *with* $\alpha \in \mathbb{R}$. *A* relative *phase sits between the terms of a superposition, like the* $e^{i\delta}$ *in* $c_+|+\hat{z}\rangle + e^{i\delta}c_-|-\hat{z}\rangle$, *and cannot be pulled out front.*

Proposition 4. *A global phase changes no probability:* $\left|\langle\phi|e^{i\alpha}\psi\rangle\right|^2 = |e^{i\alpha}|^2 |\langle\phi|\psi\rangle|^2 = |\langle\phi|\psi\rangle|^2$ *for every* $|\phi\rangle$.

Definition 25 (Physical state)**.** *What the system is actually like, as far as any experiment can tell: the probabilities of every outcome of every possible measurement. Two kets that give the same probabilities for all measurements describe the same physical state.*

Result 9. $|\psi\rangle$ *and* $e^{i\alpha}|\psi\rangle$ *are the same physical state, so global phases can be dropped. Relative phases cannot: they leave the* S_z *probabilities alone but change the* S_x *and* S_y *ones (Section 4.4).*

4.3 Expectation value and uncertainty

Definition 26 (Expectation value)**.** *The average outcome, weighted by the probabilities:*

$$\langle A\rangle = \sum_j a_j\, P(A=a_j) = \sum_j a_j\, |\langle a_j|\psi\rangle|^2\,; \qquad \langle S_z\rangle = \frac{\hbar}{2}|c_+|^2 - \frac{\hbar}{2}|c_-|^2. \quad (22)$$

Definition 27 (Uncertainty)**.** *The standard deviation (typical spread of outcomes around the average):*

$$(\Delta A)^2 = \langle (A-\langle A\rangle)^2\rangle = \langle A^2\rangle - \langle A\rangle^2\,, \qquad \langle A^2\rangle = \sum_j a_j^2\, |\langle a_j|\psi\rangle|^2\,. \quad (23)$$

Remark 13. *For spin-$\frac{1}{2}$ both outcomes square to $\hbar^2/4$, so $\langle S_z^2\rangle = \hbar^2/4$ for every state.*

Example. *For $|+\hat{x}\rangle$, Experiment 3 gives both S_z probabilities $1/2$, so $\langle S_z\rangle = 0$ and $\Delta S_z = \hbar/2$. The average is zero but every single measurement gives $\pm\hbar/2$: maximal spread.*

Example. *For $|\psi\rangle = \sqrt{\frac{1}{3}}\,|+\hat{z}\rangle + i\sqrt{\frac{2}{3}}\,|-\hat{z}\rangle$,*

$$\langle S_z\rangle = \frac{\hbar}{2}\cdot\frac{1}{3} - \frac{\hbar}{2}\cdot\frac{2}{3} = -\frac{\hbar}{6}, \qquad (\Delta S_z)^2 = \frac{\hbar^2}{4} - \frac{\hbar^2}{36} = \frac{2}{9}\hbar^2, \qquad \Delta S_z = \frac{\sqrt{2}}{3}\hbar.$$

4.3.1 Measured averages and shot noise

Remark 14. *For N measurements on identically prepared atoms, with $N_\pm$ giving $\pm\hbar/2$, the measured average is $\langle S_z\rangle_N = \frac{\hbar}{2}\frac{N_+ - N_-}{N}$, which tends to $\langle S_z\rangle$ as $N\to\infty$.*

Remark 15 (Shot noise)**.** *The measured average $\langle S_z\rangle_N$ scatters around the true expectation value, and its average deviation from it is*

$$|\langle S_z\rangle_N - \langle S_z\rangle| \sim \frac{\Delta S_z}{\sqrt{N}}.$$

This scatter is called shot noise. It comes from the randomness of individual quantum outcomes, not from a faulty apparatus, and is absent only when $\Delta S_z = 0$, i.e. when the state has a definite S_z.

4.4 Spin states along x and y

4.4.1 The state $|+\hat{x}\rangle$

Experiment 3 gives $|\,\langle\pm\hat{z}| + \hat{x}\rangle\,|^2 = \frac{1}{2}$, so both amplitudes have modulus $1/\sqrt{2}$ with their own phases:

$$|+\hat{x}\rangle = \frac{e^{i\delta_+}}{\sqrt{2}}\left(|+\hat{z}\rangle + e^{i\delta}\,|-\hat{z}\rangle\right), \qquad \delta = \delta_- - \delta_+.$$

Dropping the global phase $e^{i\delta_+}$,

$$|+\hat{x}\rangle = \frac{1}{\sqrt{2}}\left(|+\hat{z}\rangle + e^{i\delta}\,|-\hat{z}\rangle\right).$$

Experiment 3 alone does not fix the relative phase δ.

4.4.2 Experiment 5: measuring S_y

Definition 28 (Cyclic permutation)**.** *Relabelling the axes $x \to y \to z \to x$ (or the reverse). It is a rotation of the coordinate system, so no physics changes.*

Experiment 5 is Experiment 3 cyclically permuted: filter $+\hat{z}$, then measure S_y. By symmetry it also splits 50/50, so every cross-axis overlap has modulus squared $\frac{1}{2}$:

$$|\langle +\hat{x}|+\hat{z}\rangle|^2 = |\langle +\hat{y}|+\hat{z}\rangle|^2 = |\langle -\hat{x}|+\hat{y}\rangle|^2 = \frac{1}{2}.$$

The same argument as for $|+\hat{x}\rangle$ gives, after dropping global phases,

$$|\pm\hat{x}\rangle = \frac{1}{\sqrt{2}}\left(|+\hat{z}\rangle \pm e^{i\delta}\,|-\hat{z}\rangle\right), \qquad |\pm\hat{y}\rangle = \frac{1}{\sqrt{2}}\left(|+\hat{z}\rangle \pm e^{i\gamma}\,|-\hat{z}\rangle\right).$$

4.4.3 Fixing the phases

Using orthonormality of the $\hat{z}$ basis,

$$|\langle +\hat{x}|+\hat{y}\rangle|^2 = \left|\frac{1}{2}\left(1+e^{i(\gamma-\delta)}\right)\right|^2 = \frac{1}{2}\left(1+\cos(\gamma-\delta)\right).$$

Setting this to $\frac{1}{2}$ forces $\gamma - \delta = \pm\pi/2$. The rest is convention.

Result 10 (Phase convention)**.** *With $\delta = 0$ and $\gamma = +\pi/2$,*

$$|\pm\hat{x}\rangle = \frac{1}{\sqrt{2}}\left(|+\hat{z}\rangle \pm |-\hat{z}\rangle\right), \qquad |\pm\hat{y}\rangle = \frac{1}{\sqrt{2}}\left(|+\hat{z}\rangle \pm i\,|-\hat{z}\rangle\right), \tag{24}$$

$$\langle\pm\hat{x}| = \frac{1}{\sqrt{2}}\left(\langle+\hat{z}| \pm \langle-\hat{z}|\right), \qquad \langle\pm\hat{y}| = \frac{1}{\sqrt{2}}\left(\langle+\hat{z}| \mp i\,\langle-\hat{z}|\right). \tag{25}$$

Remark 16. *Complex coefficients are required: with real ones the relative phases could only be 0 or π, and $\gamma - \delta = \pm\pi/2$ would be impossible.*

Example. *For $|\psi\rangle = \sqrt{\frac{1}{3}}\,|+\hat{z}\rangle + i\sqrt{\frac{2}{3}}\,|-\hat{z}\rangle$, compute $\langle S_y\rangle$. Using $\langle\pm\hat{y}|+\hat{z}\rangle = \frac{1}{\sqrt{2}}$ and $\langle\pm\hat{y}|-\hat{z}\rangle = \mp\frac{i}{\sqrt{2}}$,*

$$\langle\pm\hat{y}|\psi\rangle = \frac{1}{\sqrt{2}}\left(\sqrt{\tfrac{1}{3}} \pm \sqrt{\tfrac{2}{3}}\right), \qquad P\big(S_y = \pm\tfrac{\hbar}{2}\big) = \frac{1}{2} \pm \frac{\sqrt{2}}{3},$$

which add to 1. Then

$$\langle S_y\rangle = \frac{\hbar}{2}\left[\left(\frac{1}{2}+\frac{\sqrt{2}}{3}\right) - \left(\frac{1}{2}-\frac{\sqrt{2}}{3}\right)\right] = \frac{\sqrt{2}}{3}\hbar.$$

The factor i did nothing for S_z, but as a relative phase it is exactly what tilts the spin toward $+\hat{y}$.

5 Matrix Mechanics

September 21-25, 2026

5.1 States as column vectors

Definition 29 (Representation of a state)**.** *In a basis, a ket is represented by the column of its coefficients and its bra by the conjugate row (the adjoint, †). In the S_z basis:*

$$|\psi\rangle \xrightarrow{S_z} \psi = \begin{pmatrix} c_+ \\ c_- \end{pmatrix}, \qquad \langle\psi| \xrightarrow{S_z} \psi^\dagger = \begin{pmatrix} c_+^* & c_-^* \end{pmatrix}, \qquad c_\pm = \langle \pm\hat{z}|\psi\rangle . \tag{26}$$

Remark 17. *Write $|\psi\rangle \xrightarrow{S_z} \psi$, not $|\psi\rangle = \psi$: the ket is basis-independent, the column is not.*

Example.

$$|+\hat{z}\rangle \xrightarrow{S_z} \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad |-\hat{z}\rangle \xrightarrow{S_z} \begin{pmatrix} 0 \\ 1 \end{pmatrix},$$

$$|\pm\hat{x}\rangle \xrightarrow{S_z} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ \pm 1 \end{pmatrix}, \qquad |\pm\hat{y}\rangle \xrightarrow{S_z} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ \pm i \end{pmatrix}.$$

Problem 2 (Other bases)**.** *In the S_x basis, $|\psi\rangle \xrightarrow{S_x} \begin{pmatrix} \langle +\hat{x}|\psi\rangle \\ \langle -\hat{x}|\psi\rangle \end{pmatrix}$, and similarly for S_y. Write the column representing $|+\hat{z}\rangle$ in the S_z, S_x and S_y bases.*

Proposition 5 (Inner product as a matrix product)**.** *If $|\phi\rangle \xrightarrow{S_z} \phi$ and $|\psi\rangle \xrightarrow{S_z} \psi$, then $\langle\phi|\psi\rangle = \phi^\dagger\psi$ (row times column), so $P(\phi\,|\,\psi) = \left|\phi^\dagger\psi\right|^2$ and $\langle\psi|\psi\rangle = \psi^\dagger\psi$.*

Remark 18. *For a state given in another basis, replace each basis ket by its S_z column and add. For $|\psi\rangle = a\,|+\hat{y}\rangle + b\,|-\hat{y}\rangle$,*

$$\psi \xrightarrow{S_z} \frac{a}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix} + \frac{b}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} a+b \\ i(a-b) \end{pmatrix}.$$

Checking $\psi^\dagger\psi = 1$ before going further catches most mistakes. A cleaner systematic method is in Section 5.4.

Problem 3 (A probability by matrix mechanics)**.** *Using matrix mechanics, find $P(S_x = +\hbar/2)$ for the state $|\psi\rangle = \sqrt{\frac{1}{5}}\,|+\hat{z}\rangle + i\sqrt{\frac{4}{5}}\,|-\hat{z}\rangle$.*

5.2 Operators as matrices

Definition 30 (Linear operator)**.** *A map A from kets to kets that respects superpositions, $A\,(a\,|\psi\rangle + b\,|\phi\rangle) = aA\,|\psi\rangle + bA\,|\phi\rangle$. Every operator in this course is linear.*

Definition 31 (Matrix of an operator)**.** *In a basis $\{|j\rangle\}$, A is represented by the matrix of* matrix elements

$$A_{ij} = \langle i|\,A\,|j\rangle . \tag{27}$$

Proposition 6 (Operator acting on a state)**.** *$A|\psi\rangle$ is represented by the matrix times the column, $A\psi$.*

Proof. The kth entry of the column for $A|\psi\rangle$ is $\langle k|A|\psi\rangle$. Inserting the identity,

$$\langle k|A|\psi\rangle = \sum_j \langle k|A|j\rangle\langle j|\psi\rangle = \sum_j A_{kj}\, c_j,$$

which is row k of A times the column ψ. □

Example. *The projectors and identity in the S_z basis:*

$$P_{+\hat{z}} \xrightarrow{S_z} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad P_{-\hat{z}} \xrightarrow{S_z} \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \qquad \mathbb{1} \xrightarrow{S_z} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix},$$

and indeed $P_{+\hat{z}}\psi = (c_+, 0)^T$.

Definition 32 (Adjoint)**.** *The adjoint $A^\dagger$ is the conjugate transpose: $(A^\dagger)_{ij} = A^*_{ji}$. In operator form, writing $A = \sum_{i,j} A_{ij}|i\rangle\langle j|$,*

$$A^\dagger = \sum_{i,j} A^*_{ij}|j\rangle\langle i|. \tag{28}$$

For real entries it is just the transpose.

Proposition 7 (Bra of a transformed state)**.** *If $|\psi'\rangle = A|\psi\rangle$, then $\langle\psi'| = \langle\psi|A^\dagger$.*

Proof. The column of $|\psi'\rangle$ is $A\psi$, and its bra is the conjugate transpose, $(A\psi)^\dagger = \psi^\dagger A^\dagger$. □

Proposition 8 (Adjoint of a product)**.**

$$(AB)^\dagger = B^\dagger A^\dagger. \tag{29}$$

The order reverses; $A^\dagger B^\dagger$ is a different matrix in general.

5.3 Observables as operators

Definition 33 (Eigenstate and eigenvalue)**.** *$|a\rangle$ is an eigenstate of A with eigenvalue a if*

$$A|a\rangle = a|a\rangle, \tag{30}$$

the familiar $Mv = \lambda v$ in matrix form. A general operator turns a state into a different state; eigenstates come back as themselves times a number. A basis made of eigenstates of A is called an eigenbasis *of A.*

Remark 19 (Eigenstate vs eigenvector)**.** *Same thing in two languages. "Eigenstate" is the physics word for the ket, e.g. $|+\hat{x}\rangle$. "Eigenvector" is the linear-algebra word, usually for its column in some basis, e.g. $\frac{1}{\sqrt{2}}(1,1)^T$ in the S_z basis.*

Proposition 9 (Operator of an observable)**.** *For an observable with outcomes a_j and states $|a_j\rangle$, define*

$$\hat{A} = \sum_j a_j \, |a_j\rangle \, \langle a_j| \, . \tag{31}$$

Then $\hat{A}\,|a_j\rangle = a_j\,|a_j\rangle$ (the outcomes are its eigenvalues, the outcome states its eigenstates), and

$$\langle A\rangle = \langle\psi|\,\hat{A}\,|\psi\rangle\,, \qquad \langle A^2\rangle = \langle\psi|\,\hat{A}^2\,|\psi\rangle\,. \tag{32}$$

Proof. $\hat{A}\,|a_k\rangle = \sum_j a_j\,|a_j\rangle\,\delta_{jk} = a_k\,|a_k\rangle$, and $\langle\psi|\,\hat{A}\,|\psi\rangle = \sum_j a_j\,\langle\psi|a_j\rangle\,\langle a_j|\psi\rangle = \sum_j a_j\,|\langle a_j|\psi\rangle|^2 = \langle A\rangle$. Likewise $\hat{A}^2 = \sum_j a_j^2\,|a_j\rangle\,\langle a_j|$. □

Proposition 10 (An operator is diagonal in its own eigenbasis)**.**

$$\langle a_i|\,\hat{A}\,|a_j\rangle = a_j\delta_{ij}, \qquad \hat{A} \xrightarrow{|a_j\rangle} D = \begin{pmatrix} a_1 & & 0 \\ & \ddots & \\ 0 & & a_n \end{pmatrix}. \tag{33}$$

5.3.1 Spin operators and Pauli matrices

Definition 34 (Spin operators)**.** *Applying the operator of an observable to spin, for $\hat{n} \in \{\hat{x}, \hat{y}, \hat{z}\}$,*

$$\hat{S}_n = \frac{\hbar}{2}\left(|+\hat{n}\rangle\,\langle+\hat{n}| - |-\hat{n}\rangle\,\langle-\hat{n}|\right), \qquad \hat{S}_n\,|\pm\hat{n}\rangle = \pm\frac{\hbar}{2}\,|\pm\hat{n}\rangle\,. \tag{34}$$

Example. *Each spin operator in its own eigenbasis, using $\hat{S}_n\,|\pm\hat{n}\rangle = \pm\frac{\hbar}{2}\,|\pm\hat{n}\rangle$ and orthonormality:*

$$\langle\pm\hat{n}|\,\hat{S}_n\,|\pm\hat{n}\rangle = \pm\frac{\hbar}{2}, \qquad \langle\pm\hat{n}|\,\hat{S}_n\,|\mp\hat{n}\rangle = \mp\frac{\hbar}{2}\,\langle\pm\hat{n}|\mp\hat{n}\rangle = 0,$$

so

$$\hat{S}_x \xrightarrow{S_x} \frac{\hbar}{2}\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}, \qquad \hat{S}_y \xrightarrow{S_y} \frac{\hbar}{2}\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}, \qquad \hat{S}_z \xrightarrow{S_z} \frac{\hbar}{2}\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}.$$

In the S_z basis the matrix elements of $\hat{S}_n$ are

$$\langle i|\,\hat{S}_n\,|j\rangle = \frac{\hbar}{2}\Big(\,\langle i| + \hat{n}\rangle\,\langle+\hat{n}|j\rangle - \langle i| - \hat{n}\rangle\,\langle-\hat{n}|j\rangle\,\Big), \qquad i,j \in \{+\hat{z}, -\hat{z}\}.$$

Problem 4 (The matrix of $\hat{S}_z$)**.** *Find the matrix representation of $\hat{S}_z$ in the S_z basis.*

Problem 5 (The matrix of $\hat{S}_x$)**.** *Find the matrix representation of $\hat{S}_x$ in the S_z basis.*

Problem 6 (The matrix of $\hat{S}_y$)**.** *Find the matrix representation of $\hat{S}_y$ in the S_z basis.*

Remark 20 (Shortcut)**.** *Write each projector as a column times its conjugate row, e.g.*

$$|\pm\hat{x}\rangle\langle\pm\hat{x}| \xrightarrow{S_z} \frac{1}{2}\begin{pmatrix}1\\ \pm1\end{pmatrix}\begin{pmatrix}1 & \pm1\end{pmatrix} = \frac{1}{2}\begin{pmatrix}1 & \pm1\\ \pm1 & 1\end{pmatrix},$$

then subtract and multiply by $\hbar/2$.

Definition 35 (Pauli matrices)**.** *The results, all in the S_z basis:*

$$\hat{S}_j = \frac{\hbar}{2}\sigma_j, \qquad \sigma_x = \begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}, \quad \sigma_y = \begin{pmatrix}0 & -i\\ i & 0\end{pmatrix}, \quad \sigma_z = \begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}. \tag{35}$$

Each equals its own adjoint and squares to $\mathbb{1}$, so $\hat{S}_j^2 = \frac{\hbar^2}{4}\mathbb{1}$. Collected into the vector of Pauli matrices $\vec{\sigma} = \sigma_x\hat{x} + \sigma_y\hat{y} + \sigma_z\hat{z}$, this reads $\vec{S} \xrightarrow{S_z} \frac{\hbar}{2}\vec{\sigma}$.

Problem 7 (Expectation value and uncertainty by matrix mechanics)**.** *For $|\psi\rangle = \sqrt{\frac{1}{5}}\,|+\hat{y}\rangle + i\sqrt{\frac{4}{5}}\,|-\hat{y}\rangle$, find $\langle S_z\rangle$ and ΔS_z using matrix mechanics.*

5.3.2 Hermitian operators

Definition 36 (Hermitian operator)**.** *A is Hermitian if it equals its own adjoint, $A = A^\dagger$ (e.g. the Pauli matrices).*

Principle 2. *Every observable is represented by a Hermitian operator.*

Proposition 11. *The eigenvalues of a Hermitian operator are real, so measurement outcomes are real.*

Proof. If $A|a\rangle = a|a\rangle$, the bra is $\langle a|A^\dagger = a^*\langle a|$, so $a\langle a|a\rangle = \langle a|A|a\rangle = \langle a|A^\dagger|a\rangle = a^*\langle a|a\rangle$, and $a = a^*$. □

5.4 Change of basis

Take an n-level system with two orthonormal bases, $\{|j\rangle\}$ and $\{|\alpha\rangle\}$, $j, \alpha = 1, \ldots, n$. Within each basis the overlaps are Kronecker deltas, but across bases they are not: $\langle+\hat{x}|+\hat{z}\rangle = \frac{1}{\sqrt{2}}$ is neither 0 nor 1. The same ket then has two columns,

$$|\psi\rangle \xrightarrow{|j\rangle} \psi^j = \begin{pmatrix}\langle 1|\psi\rangle\\ \vdots\\ \langle n|\psi\rangle\end{pmatrix}, \qquad |\psi\rangle \xrightarrow{|\alpha\rangle} \psi^\alpha = \begin{pmatrix}\langle \alpha{=}1|\psi\rangle\\ \vdots\\ \langle \alpha{=}n|\psi\rangle\end{pmatrix}.$$

Inserting $\mathbb{1} = \sum_\alpha |\alpha\rangle\langle\alpha|$ into each $|j\rangle$ amplitude,

$$\langle j|\psi\rangle = \sum_\alpha \langle j|\alpha\rangle\langle\alpha|\psi\rangle.$$

Result 11 (Change-of-basis matrix)**.**

$$\psi^j = U\psi^\alpha, \qquad \psi^\alpha = U^\dagger\psi^j, \qquad U_{j\alpha} = \langle j|\alpha\rangle\,. \tag{36}$$

The columns of U are the new basis kets $|\alpha\rangle$ written in the old basis $\{|j\rangle\}$.

Definition 37 (Unitary operator)**.** *U is unitary if $U^\dagger U = UU^\dagger = \mathbb{1}$, i.e. its inverse is its adjoint.*

Theorem 1. *The change-of-basis matrix U is unitary.*

Proof. With $(U^\dagger)_{\alpha j} = \langle j|\alpha\rangle^* = \langle\alpha|j\rangle$ and completeness of $\{|j\rangle\}$,

$$(U^\dagger U)_{\alpha\beta} = \sum_j \langle\alpha|j\rangle\langle j|\beta\rangle = \langle\alpha|\beta\rangle = \delta_{\alpha\beta}.$$

Completeness of $\{|\alpha\rangle\}$ gives $UU^\dagger = \mathbb{1}$ the same way. □

Remark 21. *So normalization does not depend on the basis: $\psi^{j\dagger}\psi^j = \psi^{\alpha\dagger}U^\dagger U\psi^\alpha = \psi^{\alpha\dagger}\psi^\alpha$.*

Remark 22. *If a unitary U maps $|\psi\rangle$ to $c\,|\phi\rangle$ with $|\phi\rangle$ normalized, then $1 = \langle\psi|\,U^\dagger U\,|\psi\rangle = |c|^2$, so $c = e^{i\varphi}$. In particular, eigenvalues of a unitary are pure phases, and U leaves a physical state unchanged only when that state is an eigenstate of U.*

Example (Between S_x and S_z)**.** *With $|j\rangle = |\pm\hat{z}\rangle$ and $|\alpha\rangle = |\pm\hat{x}\rangle$, the columns of U are the S_z columns of $|\pm\hat{x}\rangle$:*

$$U = \begin{pmatrix} \langle+\hat{z}|+\hat{x}\rangle & \langle+\hat{z}|-\hat{x}\rangle \\ \langle-\hat{z}|+\hat{x}\rangle & \langle-\hat{z}|-\hat{x}\rangle \end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}. \tag{37}$$

The two directions are

$$S_x \to S_z:\ \psi^z = U\psi^x, \qquad S_z \to S_x:\ \psi^x = U^\dagger\psi^z = U\psi^z,$$

since U is real and symmetric, so $U^\dagger = U$: the same matrix goes both ways ($U^2 = \mathbb{1}$). Bill calls U "$R\left(\frac{\pi}{2}\hat{y}\right)$", but it is not the actual rotation matrix (see the warning in Section 6.4).

Remark 23. *Between S_y and S_z the matrix has columns $\frac{1}{\sqrt{2}}(1,\pm i)^T$,*

$$U_y = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ i & -i \end{pmatrix}, \qquad \psi^z = U_y\psi^y, \qquad \psi^y = U_y^\dagger\psi^z,$$

and here $U_y^\dagger \neq U_y$, so the reverse direction needs the actual adjoint.

Problem 8 (Expectation value from the S_x basis)**.** *For $|\psi\rangle = \frac{1}{\sqrt{3}}\,|+\hat{x}\rangle + i\sqrt{\frac{2}{3}}\,|-\hat{x}\rangle$, find $\langle S_z\rangle = \psi^{z\dagger}\hat{S}_z\psi^z$ using matrix mechanics.*

5.4.1 Operators in another basis

Let $\{|j\rangle\} = \{|a_j\rangle\}$ be the eigenbasis of $\hat{A}$, so $\hat{A} \xrightarrow{|a_j\rangle} D$ (diagonal), and $U_{j\alpha} = \langle a_j|\alpha\rangle$.

Proposition 12 (Similarity transformation)**.**

$$\hat{A} \xrightarrow{|\alpha\rangle} A^{\alpha} = U^{\dagger} D\, U. \tag{38}$$

Rewriting the same operator in a new basis this way is called a similarity transformation.

Proof. Insert the identity twice:

$$\langle\alpha|\,\hat{A}\,|\beta\rangle = \sum_{i,j} \langle\alpha|a_i\rangle\,\langle a_i|\,\hat{A}\,|a_j\rangle\,\langle a_j|\beta\rangle = \sum_{i,j} (U^{\dagger})_{\alpha i} D_{ij} U_{j\beta} = (U^{\dagger} D\, U)_{\alpha\beta}.$$

□

Result 12 (Expectation value in any basis)**.**

$$\langle A\rangle = \psi^{j\dagger} D\, \psi^{j} = \psi^{\alpha\dagger} \left[U^{\dagger} D\, U\right] \psi^{\alpha}. \tag{39}$$

Either move the state to the eigenbasis, or move the operator to the state's basis. Both give the same number.

Problem 9 ($\hat{S}_z$ in the S_x basis)**.** *Using U from Eq. (37), find the matrix of $\hat{S}_z$ in the S_x basis, and use it to redo the problem "Expectation value from the S_x basis" without changing the state's basis.*

6 Rotations

September 21-28, 2026

6.1 The rotation operator

Definition 38 (Rotation operator)**.** *$R(\theta\hat{n})$ is the linear operator that rotates a state by angle θ about the axis $\hat{n}$: $|\psi'\rangle = R(\theta\hat{n})\,|\psi\rangle$.*

Example. *$R\left(\frac{\pi}{2}\hat{y}\right)$ turns $\hat{z}$ into $\hat{x}$, so it takes $|\pm\hat{z}\rangle$ to $|\pm\hat{x}\rangle$ up to phase, and by linearity*

$$R\left(\tfrac{\pi}{2}\hat{y}\right)\left(c_+\,|+\hat{z}\rangle + c_-\,|-\hat{z}\rangle\right) = c_+\,|+\hat{x}\rangle + c_-\,|-\hat{x}\rangle\,.$$

Up to a phase on one column, its S_z matrix is the change-of-basis matrix U of Eq. (37) (exact form in Section 6.4).

Remark 24 (Why the bra carries $R^{\dagger}$)**.** *By the bra-of-a-transformed-state proposition, $\langle\psi'| = \langle\psi|\,R^{\dagger}$. Using $\langle\psi|\,R$ instead fails: with $|+\hat{x}\rangle = R\left(\frac{\pi}{2}\hat{y}\right)|+\hat{z}\rangle$ it would give*

$$\langle+\hat{x}|+\hat{x}\rangle = \langle+\hat{z}|\,R\left(\tfrac{\pi}{2}\hat{y}\right)R\left(\tfrac{\pi}{2}\hat{y}\right)|+\hat{z}\rangle = \langle+\hat{z}|\,R(\pi\hat{y})\,|+\hat{z}\rangle \propto \langle+\hat{z}|-\hat{z}\rangle = 0,$$

since a π rotation about $\hat{y}$ flips $+\hat{z}$ to $-\hat{z}$. That contradicts $\langle+\hat{x}|+\hat{x}\rangle = 1$.

Theorem 2. *Rotation operators are unitary: $R^{\dagger}R = RR^{\dagger} = \mathbb{1}$.*

Proof. A rotation must preserve normalization: $\langle\psi'|\psi'\rangle = \langle\psi|\,R^{\dagger}R\,|\psi\rangle = 1$ for every $|\psi\rangle$, which forces $R^{\dagger}R = \mathbb{1}$. □

6.2 Generators of rotations

Rotations about the same axis add, $R(\theta_1\hat{z})R(\theta_2\hat{z}) = R\big((\theta_1+\theta_2)\hat{z}\big)$, so a finite rotation is $N \gg 1$ small ones,

$$R(\theta\hat{z}) = \left[R\left(\tfrac{\theta}{N}\hat{z}\right)\right]^N. \tag{40}$$

Taylor expanding one small step about the identity and naming the first-order operator $-iJ_z/\hbar$,

$$R\left(\tfrac{\theta}{N}\hat{z}\right) = \mathbb{1} - \frac{i}{\hbar}J_z\frac{\theta}{N} + \mathcal{O}\left(\frac{1}{N^2}\right). \tag{41}$$

Result 13. *By the problem "The exponential as a limit",*

$$R(\theta\hat{z}) = \lim_{N\to\infty}\left(\mathbb{1} - \frac{i}{\hbar}J_z\frac{\theta}{N}\right)^N = e^{-iJ_z\theta/\hbar}, \tag{42}$$

and likewise $R(\theta\hat{n}) = e^{-iJ_n\theta/\hbar}$ for any axis.

Definition 39 (Generator of rotations)**.** *The operator J_z above: the thing that, exponentiated, produces rotations about $\hat{z}$. It has units of $\hbar$, i.e. of angular momentum ($[\hbar] = [\vec{r}\times\vec{p}]$), so $\vec{J}$ is identified as the angular momentum operator. For spin-$\frac{1}{2}$, $\vec{J} = \vec{S}$ and $R(\theta\hat{n}) = e^{-i\theta\hat{S}_n/\hbar}$.*

Proposition 13. *The generator J_z is Hermitian (Def. 36).*

Proof. To first order in θ, $R = \mathbb{1} - i\theta J_z/\hbar$ and $R^\dagger = \mathbb{1} + i\theta J_z^\dagger/\hbar$. Unitarity gives

$$\mathbb{1} = R^\dagger R = \mathbb{1} + \frac{i\theta}{\hbar}\left(J_z^\dagger - J_z\right) + \mathcal{O}(\theta^2),$$

so $J_z^\dagger = J_z$. □

Remark 25. *So J_z qualifies as an observable (Principle 2), consistent with identifying it as angular momentum.*

6.3 Functions of an operator

Proposition 14 (Function of an operator on an eigenstate)**.** *If $\hat{A}\,|a\rangle = a\,|a\rangle$ and f has a Taylor series, then*

$$f(\hat{A})\,|a\rangle = f(a)\,|a\rangle. \tag{43}$$

Proof. $f(\hat{A})$ is defined by its Taylor series, and $\hat{A}^n\,|a\rangle = a^n\,|a\rangle$, so term by term

$$f(\hat{A})\,|a\rangle = \sum_n \frac{f^{(n)}(0)}{n!}\hat{A}^n\,|a\rangle = \sum_n \frac{f^{(n)}(0)}{n!}a^n\,|a\rangle = f(a)\,|a\rangle.$$

□

Proposition 15 (Spectral form)**.** *Writing an operator as a sum of its eigenvalues times the projectors onto its eigenstates is its spectral form. Inserting* $\mathbb{1} = \sum_j |a_j\rangle \langle a_j|$ *and using Proposition 14,*

$$f(\hat{A}) = \sum_j f(a_j) |a_j\rangle \langle a_j| . \tag{44}$$

With $f(x) = x$ *this is the observable operator* $\hat{A} = \sum_j a_j |a_j\rangle \langle a_j|$.

6.3.1 Rotations about $\hat{z}$

Result 14. *With* $f(x) = e^{-i\phi x/\hbar}$ *and* $\hat{S}_z |\pm\hat{z}\rangle = \pm\frac{\hbar}{2} |\pm\hat{z}\rangle$,

$$R(\phi\hat{z}) |\pm\hat{z}\rangle = e^{\mp i\phi/2} |\pm\hat{z}\rangle , \qquad R(\phi\hat{z}) \xrightarrow{S_z} \begin{pmatrix} e^{-i\phi/2} & 0 \\ 0 & e^{i\phi/2} \end{pmatrix} . \tag{45}$$

This makes sense: a rotation about $\hat{z}$ *cannot change* S_z*, so it can only attach phases to the* S_z *states. In other bases the matrix is not diagonal.*

Remark 26. *A full turn,* $\phi = 2\pi$*, gives* $R = -\mathbb{1}$*: a spin-*$\frac{1}{2}$ *state needs* 4π *to return to itself. The sign is a global phase, so it is invisible on its own. Its consequences for many particles are in Section 7.7.*

Problem 10 ($R(\phi\hat{z})$ in the S_x basis)**.** *Find the matrix representation of* $R(\phi\hat{z})$ *in the* S_x *basis.*

6.4 Rotations about any axis

Proposition 16 (Spectral form of a rotation)**.** *For* $\hat{n} \in \{\hat{x}, \hat{y}, \hat{z}\}$,

$$R(\phi\hat{n}) = e^{-i\phi/2} |+\hat{n}\rangle \langle +\hat{n}| + e^{i\phi/2} |-\hat{n}\rangle \langle -\hat{n}| . \tag{46}$$

Proof. Spectral form with $f(x) = e^{-i\phi x/\hbar}$, $\hat{A} = \hat{S}_n$, eigenvalues $\pm\hbar/2$. □

Remark 27 (Recipe)**.** *In any basis* $\{|i\rangle\}$ *this reads* $VDV^\dagger$*, with* $D = \operatorname{diag}\left(e^{-i\phi/2}, e^{i\phi/2}\right)$ *and* V *the matrix whose columns are* $|\pm\hat{n}\rangle$ *written in* $\{|i\rangle\}$*. It is the similarity transformation with the roles of the two bases swapped.*

Proposition 17 ($R(\phi\hat{x})$ in the S_z basis)**.**

$$R(\phi\hat{x}) \xrightarrow{S_z} \begin{pmatrix} \cos\frac{\phi}{2} & -i\sin\frac{\phi}{2} \\ -i\sin\frac{\phi}{2} & \cos\frac{\phi}{2} \end{pmatrix} . \tag{47}$$

Proof. Here $V = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = V^\dagger$, so

$$R(\phi\hat{x}) \xrightarrow{S_z} \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} e^{-i\phi/2} & 0 \\ 0 & e^{i\phi/2} \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} e^{-i\phi/2} + e^{i\phi/2} & e^{-i\phi/2} - e^{i\phi/2} \\ e^{-i\phi/2} - e^{i\phi/2} & e^{-i\phi/2} + e^{i\phi/2} \end{pmatrix} ,$$

and Euler's formula gives $2\cos\frac{\phi}{2}$ and $-2i\sin\frac{\phi}{2}$. □

Remark 28. *This is the same matrix as $R(\phi\hat{z})$ in the S_x basis. It is also the same as $R(\phi\hat{z})$ in the S_y basis (PS2 Q8): with $|\pm\hat{y}\rangle = \frac{1}{\sqrt{2}}(|+\hat{z}\rangle \pm i\,|-\hat{z}\rangle)$,*

$$\langle\pm\hat{y}|\, R(\phi\hat{z})\, |\pm\hat{y}\rangle = \cos\frac{\phi}{2}, \qquad \langle\pm\hat{y}|\, R(\phi\hat{z})\, |\mp\hat{y}\rangle = -i\sin\frac{\phi}{2},$$

so in both bases

$$R(\phi\hat{z}) \longrightarrow \begin{pmatrix} \cos\frac{\phi}{2} & -i\sin\frac{\phi}{2} \\ -i\sin\frac{\phi}{2} & \cos\frac{\phi}{2} \end{pmatrix}.$$

Proposition 18 ($R(\phi\hat{y})$ in the S_z basis)**.**

$$R(\phi\hat{y}) \xrightarrow{S_z} \begin{pmatrix} \cos\frac{\phi}{2} & -\sin\frac{\phi}{2} \\ \sin\frac{\phi}{2} & \cos\frac{\phi}{2} \end{pmatrix}. \tag{48}$$

Proof. Now $V = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ i & -i \end{pmatrix}$ and $V^\dagger = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & -i \\ 1 & i \end{pmatrix} \neq V$. Then

$$R(\phi\hat{y}) \xrightarrow{S_z} \frac{1}{2}\begin{pmatrix} e^{-i\phi/2} + e^{i\phi/2} & -i\left(e^{-i\phi/2} - e^{i\phi/2}\right) \\ i\left(e^{-i\phi/2} - e^{i\phi/2}\right) & e^{-i\phi/2} + e^{i\phi/2} \end{pmatrix},$$

and with $e^{-i\phi/2} - e^{i\phi/2} = -2i\sin\frac{\phi}{2}$ the off-diagonal entries are $\mp\sin\frac{\phi}{2}$. □

Remark 29. *This matrix is real, so $R(\phi\hat{y})$ never introduces factors of i into an S_z column. At $\phi = \pi$ it gives $R(\pi\hat{y})\,|+\hat{z}\rangle = |-\hat{z}\rangle$, the flip used above to show bras carry $R^\dagger$.*

Remark 30 (WARNING: rotation versus change of basis)**.** *Plugging $\phi = \pi/2$ into this proposition does* not *give Bill's change-of-basis matrix:*

$$R\left(\tfrac{\pi}{2}\hat{y}\right) \xrightarrow{S_z} \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \neq U = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

The true rotation sends $|-\hat{z}\rangle$ to $-\,|-\hat{x}\rangle$, not $|-\hat{x}\rangle$, so its second column is $-\,|-\hat{x}\rangle$ while U in Eq. (37) has $+\,|-\hat{x}\rangle$. They differ only by the phase of one basis ket, which is a free choice, so both are valid. For a change of basis with our convention for $|\pm\hat{x}\rangle$, always use U from Eq. (37).

7 Angular Momentum

September 28, 2026

7.1 Non-commuting rotations and commutation relations

Definition 40 (Commutator)**.**

$$[A, B] = AB - BA. \tag{49}$$

A and B commute if $[A, B] = 0$, i.e. the order in which they act does not matter. Directly from the definition, $[B, A] = -[A, B]$ and $[A, BC] = B[A, C] + [A, B]C$.

Rotations about different axes do not commute: $R(\theta\hat{x})R(\theta\hat{y}) \neq R(\theta\hat{y})R(\theta\hat{x})$.

Remark 31. *Rotate a textbook by 90° about $\hat{x}$ then about $\hat{y}$ (right-hand rule). Repeat from the same start with $\hat{y}$ first. The spine ends up pointing in different directions.*

Remark 32 (Why the generators cannot commute)**.** *With $R_x = R(\theta\hat{x}) = e^{-i\theta J_x/\hbar}$ and $R_y = R(\theta\hat{y}) = e^{-i\theta J_y/\hbar}$, one might guess $R_xR_y = e^{-i\theta(J_x+J_y)/\hbar}$. That would force $R_xR_y = R_yR_x$, since $J_x + J_y = J_y + J_x$, which the textbook shows is false. The guess fails because*

$$e^A e^B = e^{A+B} \quad \text{if } [A,B] = 0, \tag{50}$$

but not in general otherwise. So the non-commutativity of rotations tells us $[J_x, J_y] = J_xJ_y - J_yJ_x \neq 0$.

Example ($[S_x, S_y]$ for spin-$\frac{1}{2}$)**.** *With $\hat{S}_j \xrightarrow{S_z} \frac{\hbar}{2}\sigma_j$,*

$$[S_x,S_y] \xrightarrow{S_z} \left(\frac{\hbar}{2}\right)^2(\sigma_x\sigma_y - \sigma_y\sigma_x) = \left(\frac{\hbar}{2}\right)^2\left[\begin{pmatrix} i & 0 \\ 0 & -i\end{pmatrix} - \begin{pmatrix} -i & 0 \\ 0 & i\end{pmatrix}\right] = i\hbar\,\frac{\hbar}{2}\sigma_z \xleftarrow{S_z} i\hbar S_z,$$

so $[S_x, S_y] = i\hbar S_z$.

Result 15 (Angular momentum commutation relations)**.** *The same relation holds for any angular momentum, and cyclic permutations ($x \to y \to z \to x$) give the other two:*

$$[J_x,J_y] = i\hbar J_z, \qquad [J_y,J_z] = i\hbar J_x, \qquad [J_z,J_x] = i\hbar J_y, \tag{51}$$

or compactly $\vec{J}\times\vec{J} = i\hbar\vec{J}$, with $\vec{J} = J_x\hat{x} + J_y\hat{y} + J_z\hat{z}$ and $\times$ the vector cross product (e.g. $\hat{x}\times\hat{y} = \hat{z}$).

7.2 Compatible observables

Definition 41 (Degenerate eigenvalue)**.** *An eigenvalue is degenerate if more than one independent eigenstate has it. Those eigenstates span a* subspace*: a smaller vector space inside the full one, closed under linear combinations.*

Theorem 3. *If $[A, B] = 0$, then A and B have a common eigenbasis.*

Proof. Let $A|a\rangle = a|a\rangle$. Since $AB = BA$,

$$A\big(B|a\rangle\big) = BA|a\rangle = a\big(B|a\rangle\big),$$

so $B|a\rangle$ is also an eigenstate of A with eigenvalue a.

Nondegenerate case. If $|a\rangle$ is the only state with eigenvalue a, then $B|a\rangle = b|a\rangle$, so $|a\rangle$ is an eigenstate of both.

Degenerate case. If n states $|a,i\rangle$ share eigenvalue a, then B maps their subspace into itself, so there it is an $n\times n$ matrix $B_{ij}=\langle a,i|\,B\,|a,j\rangle$. Diagonalizing it (finding the basis in which it is diagonal) gives eigenstates

$$|a,b_j\rangle=\sum_i |a,i\rangle\,\langle a,i|a,b_j\rangle\,,\qquad A\,|a,b_j\rangle=a\,|a,b_j\rangle\,,\qquad B\,|a,b_j\rangle=b_j\,|a,b_j\rangle\,,$$

which are still eigenstates of A because they are combinations of states with the same eigenvalue a. □

Remark 33 (Block-diagonal picture)**.** *Why B cannot mix different eigenvalues of A: if $B\,|a,i\rangle=\sum_{a',j}\langle a',j|B|a,i\rangle\,|a',j\rangle$ contained states with $a'\neq a$, then*

$$A\big(B\,|a,i\rangle\,\big)=\sum_{a',j}\langle a',j|B|a,i\rangle\;a'\,|a',j\rangle\,,$$

which equals $a\big(B\,|a,i\rangle\,\big)$ only if $\langle a',j|B|a,i\rangle\propto\delta_{aa'}$. So in the basis $\{|a_i,j\rangle\}$, B is block diagonal, with one block per eigenvalue of A:

$$B\xrightarrow{\{|a_i,j\rangle\}}\begin{pmatrix}B(a_1) & 0 & \cdots\\ 0 & B(a_2) & \\ \vdots & & \ddots\end{pmatrix},\qquad B_{kl}(a_i)=\langle a_i,k|B|a_i,l\rangle\,.$$

Diagonalizing each block separately gives the simultaneous eigenstates, $|a,b_j\rangle=\sum_i\langle a,i|a,b_j\rangle\,|a,i\rangle$.

Definition 42 (Compatible observables)**.** *Observables that commute, and so can have definite values at the same time. Otherwise they are incompatible.*

Remark 34. *$[J_x,J_y]=i\hbar J_z\neq 0$, so the components of angular momentum are incompatible: they have no common eigenbasis (for spin-$\frac{1}{2}$, no common eigenstate at all). This is the operator version of Experiment 3: putting $|+\hat z\rangle$ into an SGx gives a 50/50 mix of $|\pm\hat x\rangle$, so no state gives $S_z=+\hbar/2$ and $S_x=+\hbar/2$ with certainty. Preparing an eigenstate of one observable leaves the outcomes of an incompatible one uncertain; for $|+\hat z\rangle$ and S_x the uncertainty is maximal, $\Delta S_x=\hbar/2$.*

7.3 Uncertainty relations

Result 16 (Generalized uncertainty relation)**.** *For observables A and B, in any state,*

$$\Delta A\,\Delta B\geq\frac{1}{2}\,|\langle[A,B]\rangle|\,. \tag{52}$$

The derivation is in Townsend, Eq. (3.74). The bound only says something when $[A,B]\neq 0$.

Result 17. *For angular momentum, $[J_x,J_y]=i\hbar J_z$ gives*

$$\Delta J_x\,\Delta J_y\geq\frac{\hbar}{2}\,|\langle J_z\rangle|\,. \tag{53}$$

Example. *For $|+\hat{z}\rangle$, $\langle S_z\rangle = \hbar/2$, and both S_x and S_y give $\pm\hbar/2$ with probability $\frac{1}{2}$, so $\Delta S_x = \Delta S_y = \hbar/2$. Then $\Delta S_x \, \Delta S_y = \hbar^2/4 = \frac{\hbar}{2}\langle S_z\rangle$: the bound is saturated.*

Remark 35. *Position and momentum will also turn out not to commute, $[x, p_x] = i\hbar$, which gives the Heisenberg uncertainty relation*

$$\Delta x \, \Delta p_x \geq \frac{\hbar}{2}. \tag{54}$$

7.4 Total angular momentum

Definition 43 (Total angular momentum)**.** *The generator $\vec{J}$ of Section 6.2 is the total angular momentum, the sum of the orbital and spin angular momenta:*

$$\vec{J} = \vec{L} + \vec{S}. \tag{55}$$

Definition 44 (Square of the angular momentum)**.**

$$\vec{J}^2 = J_x^2 + J_y^2 + J_z^2. \tag{56}$$

Proposition 19. *$\vec{J}^2$ commutes with every component of $\vec{J}$; in particular $[\vec{J}^2, J_z] = 0$.*

Proof. Using $[A, BC] = B[A, C] + [A, B]C$:

$$\begin{aligned}
[J_z, J_x^2] &= J_x[J_z, J_x] + [J_z, J_x]J_x = i\hbar\,(J_xJ_y + J_yJ_x)\,, \\
[J_z, J_y^2] &= J_y[J_z, J_y] + [J_z, J_y]J_y = -i\hbar\,(J_yJ_x + J_xJ_y)\,, \\
[J_z, J_z^2] &= 0,
\end{aligned}$$

using $[J_z, J_y] = -i\hbar J_x$. The first two cancel, so $[J_z, \vec{J}^2] = 0$. The same steps work for x and y. □

Remark 36. *$\vec{J}^2$ and J_z are compatible, so they share an eigenbasis.*

Definition 45 (Angular momentum eigenstates)**.** *The simultaneous eigenstates of $\vec{J}^2$ and J_z are labelled $|j, m\rangle$, with $\langle j, m|j, m'\rangle = \delta_{mm'}$. By convention their eigenvalues are written*

$$\vec{J}^2\,|j, m\rangle = \hbar^2 j(j+1)\,|j, m\rangle\,, \qquad J_z\,|j, m\rangle = \hbar m\,|j, m\rangle\,. \tag{57}$$

For spin, $j = s$, matching the labels $|s, m\rangle$ and the magnitude $|\vec{S}| = \hbar\sqrt{s(s+1)}$ used earlier. The allowed values of j and m are derived in Section 7.5.

7.5 Ladder operators

Definition 46 (Ladder operators)**.**

$$J_\pm = J_x \pm iJ_y, \qquad J_x = \frac{1}{2}\left(J_+ + J_-\right), \qquad J_y = \frac{1}{2i}\left(J_+ - J_-\right). \tag{58}$$

They are not Hermitian: $J_+^\dagger = J_x - iJ_y = J_-$.

Proposition 20.

$$[J_z, J_\pm] = \pm\hbar J_\pm. \tag{59}$$

Proof. $[J_z, J_\pm] = [J_z, J_x] \pm i[J_z, J_y] = i\hbar J_y \pm i(-i\hbar J_x) = \pm\hbar\left(J_x \pm iJ_y\right)$. □

Proposition 21. $J_\pm \left|j,m\right\rangle$ *is an eigenstate of* J_z *with eigenvalue* $\hbar(m \pm 1)$*, so* $J_\pm \left|j,m\right\rangle \propto \left|j,m\pm1\right\rangle$.

Proof. From $[J_z, J_-] = -\hbar J_-$, $J_zJ_- = J_-J_z - \hbar J_- = J_-(J_z - \hbar)$, so

$$J_z\big(J_-\left|j,m\right\rangle\big) = J_-(J_z - \hbar)\left|j,m\right\rangle = \hbar(m-1)\big(J_-\left|j,m\right\rangle\big).$$

The same steps with $J_zJ_+ = J_+(J_z+\hbar)$ give $\hbar(m+1)$. Since $\vec{J}^2$ commutes with J_x and J_y, it commutes with $J_\pm$, so j is unchanged. □

Proposition 22 (Action of the ladder operators)**.**

$$J_\pm \left|j,m\right\rangle = \hbar\sqrt{j(j+1) - m(m\pm1)}\,\left|j,m\pm1\right\rangle. \tag{60}$$

J_- *is the* lowering operator *and* J_+ *the* raising operator.

Proof. Since $J_+ = J_-^\dagger$, the squared norm of $J_-\left|j,m\right\rangle$ is $\langle j,m|J_+J_-|j,m\rangle$. Expanding,

$$J_+J_- = (J_x + iJ_y)(J_x - iJ_y) = J_x^2 + J_y^2 - i[J_x, J_y] = \vec{J}^2 - J_z^2 + \hbar J_z,$$

so with Eq. (57),

$$\langle j,m|J_+J_-|j,m\rangle = \hbar^2\left[j(j+1) - m^2 + m\right] = \hbar^2\left[j(j+1) - m(m-1)\right].$$

Taking the coefficient real and positive (a phase convention) gives the J_- result. The same steps with $J_-J_+ = \vec{J}^2 - J_z^2 - \hbar J_z$ give the J_+ result. □

Result 18 (Allowed values of j and m)**.** *The ladder stops at both ends:*

$$J_+\left|j,j\right\rangle = \hbar\sqrt{j(j+1) - j(j+1)}\,\left|j,j+1\right\rangle = 0, \qquad J_-\left|j,-j\right\rangle = \hbar\sqrt{j(j+1) - j(j+1)}\,\left|j,-j-1\right\rangle = 0.$$

Starting from $m = j$ *and lowering,* $J_-\left|j,j\right\rangle \propto \left|j,j-1\right\rangle$, $J_-^2\left|j,j\right\rangle \propto \left|j,j-2\right\rangle$, $\ldots$, $J_-^{2j}\left|j,j\right\rangle \propto \left|j,-j\right\rangle$. *Reaching* $-j$ *in whole steps requires* $2j$ *to be a whole number, so*

$$j = 0, \tfrac{1}{2}, 1, \tfrac{3}{2}, \ldots, \qquad m = -j, -j+1, \ldots, j-1, j, \tag{61}$$

giving $2j+1$ *states for each* j*, as in Eq.* (15).

Remark 37. *The ladder cannot run past these ends because squared norms are non-negative:* $j(j+1) - m(m\pm1) \geq 0$ *keeps* $-j \leq m \leq j$.

7.6 Matrix elements

Result 19. *In the* $\{|j,m\rangle\}$ *basis,*

$$\langle j,m'|J_\pm|j,m\rangle = \hbar\sqrt{j(j+1)-m(m\pm1)}\,\delta_{m',m\pm1}, \qquad \langle j,m'|J_z|j,m\rangle = \hbar m\,\delta_{m'm}. \tag{62}$$

Example (Spin 1)**.** *For* $j=1$ *there are three states,* $m=1,0,-1$ *(e.g. the deuteron). Ordering the basis as* $\{|1,1\rangle, |1,0\rangle, |1,-1\rangle\}$*,*

$$J_z \longrightarrow \hbar\begin{pmatrix}1&0&0\\0&0&0\\0&0&-1\end{pmatrix}, \qquad J_+ \longrightarrow \hbar\begin{pmatrix}0&\sqrt{2}&0\\0&0&\sqrt{2}\\0&0&0\end{pmatrix}, \qquad J_- \longrightarrow \hbar\begin{pmatrix}0&0&0\\\sqrt{2}&0&0\\0&\sqrt{2}&0\end{pmatrix}.$$

With $J_x=\frac{1}{2}(J_+ + J_-)$ *and* $J_y=\frac{1}{2i}(J_+ - J_-)$*,*

$$J_x \longrightarrow \frac{\hbar}{\sqrt{2}}\begin{pmatrix}0&1&0\\1&0&1\\0&1&0\end{pmatrix}, \qquad J_y \longrightarrow \frac{i\hbar}{\sqrt{2}}\begin{pmatrix}0&-1&0\\1&0&-1\\0&1&0\end{pmatrix}.$$

Check $[J_x,J_y]=i\hbar J_z$*:*

$$[J_x,J_y] \longrightarrow \frac{i\hbar^2}{2}\left[\begin{pmatrix}1&0&-1\\0&0&0\\1&0&-1\end{pmatrix}-\begin{pmatrix}-1&0&-1\\0&0&0\\1&0&1\end{pmatrix}\right] = i\hbar\cdot\hbar\begin{pmatrix}1&0&0\\0&0&0\\0&0&-1\end{pmatrix} \longleftarrow i\hbar J_z.$$

7.7 Rotations by 2π: fermions and bosons

Using the spectral form (Section 6.4) with j fixed,

$$R(2\pi\hat{z}) = e^{-2\pi i J_z/\hbar}\sum_m |j,m\rangle\langle j,m| = \sum_m e^{-2\pi i m}\,|j,m\rangle\langle j,m|\,.$$

Result 20. $R(2\pi\hat{z})=\mathbb{1}$ *only if every* m *is an integer, i.e.* $j=0,1,2,\ldots$*. For half-integer* j*, every* m *is a half-integer and* $R(2\pi\hat{z})=-\mathbb{1}$*.*

Remark 38. *If we demanded that a* 2π *rotation leave every state unchanged,* j *would have to be an integer. But electrons have* $j=\frac{1}{2}$*, with* $m=\pm\frac{1}{2}$ *and* $S_z=\pm\hbar/2$*, so* $R(2\pi\hat{z})=e^{\mp i\pi}=-\mathbb{1}$ *(Remark 26). For a single spin-*$\frac{1}{2}$ *particle this is a global phase,* $R(2\pi\hat{z})\,|\psi\rangle = -\,|\psi\rangle = e^{i\pi}\,|\psi\rangle$*, with no effect on observables. For several particles, this sign for half-integer spin results in the Pauli exclusion principle.*

Definition 47 (Fermions and bosons)**.** *Particles with half-integer spin* $j=\frac{1}{2},\frac{3}{2},\frac{5}{2},\ldots$ *are* fermions *and obey Pauli exclusion. Particles with integer spin* $j=0,1,2,\ldots$ *are* bosons*.*

Remark 39. *Loosely, this connection to rotations is related to the* exchange *of two identical particles: swapping them is like rotating one around the other.*

7.8 Atomic quantum numbers

Definition 48 (Atomic quantum numbers)**.** *An electron in an atom is labelled* $|n, l, m_l, s, m_s\rangle$*, where*

- $n = 1, 2, 3, \ldots$ *is the* principal *quantum number: the size and energy of the electron cloud.*
- $l = 0, 1, \ldots, n-1$ *is the* azimuthal *quantum number: the shape of the electron cloud, with* $|\vec{L}| = \hbar\sqrt{l(l+1)}$*. The values* $l = 0, 1, 2, 3$ *are called s, p, d, f.*
- $m_l = -l, -l+1, \ldots, +l$ *is the* magnetic *quantum number: the spatial orientation of the electron cloud, with* $L_z = m_l\hbar$*.*
- $s = \frac{1}{2}$ *is the* spin *quantum number: the intrinsic angular momentum of the electron, with* $m_s = \pm\frac{1}{2}$ *giving* $S_z = m_s\hbar$*.*

The total angular momentum $\vec{J} = \vec{L} + \vec{S}$ *has its own quantum number* j*, running from* $|l - s|$ *to* $l + s$ *in steps of 1. For an electron,* $j = l \pm \frac{1}{2}$ *(only* $j = \frac{1}{2}$ *when* $l = 0$*).*

Remark 40. *No two electrons in an atom share all of* n*,* l*,* m_l*,* m_s *(Pauli exclusion principle). A subshell* l *has* $2l + 1$ *values of* m_l*, each with two values of* m_s*, so it holds* $2(2l + 1)$ *electrons: 2 for s, 6 for p, 10 for d, 14 for f.*

Remark 41. *Two notation clashes: the subshell name s (*$l = 0$*) is not the spin quantum number s, and the quantum number* j *is not the outcome label* j *in* $|j\rangle$*.*

8 Time Evolution

September 30 - October 16, 2026

8.1 The time-evolution operator

A small rotation is generated by angular momentum (Section 6.2),

$$R(d\phi\,\hat{n}) = \mathbb{1} - \frac{i}{\hbar}\left(\vec{J}\cdot\hat{n}\right)d\phi, \qquad R(\phi\hat{n})\,|\psi_0\rangle = |\psi_\phi\rangle\,,$$

where ϕ is the parameter that labels how far the state has been rotated. Time evolution works the same way, with t as the parameter.

Definition 49 (Time-evolution operator)**.** *The linear operator* $U(t)$ *that takes a state from time* 0 *to time* t*:*

$$|\psi(t)\rangle = U(t)\,|\psi(0)\rangle\,. \tag{63}$$

Remark 42. *Not to be confused with the change-of-basis matrix* U *of Section 5.4. Both are unitary, which is where the shared letter comes from.*

Proposition 23. $U(t)$ *is unitary:* $U^\dagger(t)U(t) = \mathbb{1}$*.*

Proof. Probabilities must still add to 1 at every time,

$$\langle\psi(t)|\psi(t)\rangle = \langle\psi(0)|\, U^\dagger(t)U(t)\,|\psi(0)\rangle = \langle\psi(0)|\psi(0)\rangle = 1$$

for every $|\psi(0)\rangle$, which forces $U^\dagger U = \mathbb{1}$. □

Definition 50 (Hamiltonian)**.** *The generator of time evolution, defined by the first-order expansion*

$$U(dt) = \mathbb{1} - \frac{i}{\hbar} H\, dt. \tag{64}$$

It has units $[H] = [\hbar/t]$ = energy, so H is identified as the energy operator.

Remark 43. *The same argument as for J_z (Section 6.2) shows that unitarity of U forces $H = H^\dagger$, so energy is an observable (Principle 2).*

Proposition 24 (Schrödinger equation)**.**

$$\frac{d}{dt}\,|\psi(t)\rangle = -\frac{i}{\hbar} H\,|\psi(t)\rangle\,. \tag{65}$$

Proof. Two evolutions in a row compose,

$$U(t+\Delta t) = U(\Delta t)U(t) = \left(\mathbb{1} - \frac{i}{\hbar}H\Delta t + \mathcal{O}(\Delta t^2)\right) U(t),$$

so

$$\frac{dU}{dt} = \lim_{\Delta t\to 0} \frac{U(t+\Delta t) - U(t)}{\Delta t} = -\frac{i}{\hbar} HU(t).$$

Acting on $|\psi(0)\rangle$ gives the result. □

Result 21 (Time-independent Hamiltonian)**.** *If H does not depend on time,*

$$U(t) = e^{-iHt/\hbar}. \tag{66}$$

Check: $\frac{d}{dt}e^{-iHt/\hbar} = -\frac{i}{\hbar}He^{-iHt/\hbar}$, and the initial condition $U(0) = e^0 = \mathbb{1}$ holds.

8.2 Stationary states

Definition 51 (Stationary state)**.** *A state that does not evolve into a physically different state. These are the eigenstates of H, the energy eigenstates $H\,|E\rangle = E\,|E\rangle$.*

Proposition 25.

$$U(t)\,|E\rangle = e^{-iEt/\hbar}\,|E\rangle\,. \tag{67}$$

This is a global phase, so $|E\rangle$ and $U(t)\,|E\rangle$ are the same physical state for all t.

Proof. Proposition 14 with $f(x) = e^{-ixt/\hbar}$. □

Result 22 (Evolution of a general state)**.** *Inserting the identity in the energy eigenbasis,* $|\psi(0)\rangle = \sum_n |E_n\rangle \langle E_n|\psi(0)\rangle$*, and evolving each term,*

$$|\psi(t)\rangle = \sum_n \langle E_n|\psi(0)\rangle \; e^{-iE_n t/\hbar} \, |E_n\rangle \, . \tag{68}$$

Recipe: expand in energy eigenstates, then attach $e^{-iE_n t/\hbar}$ *to each. The phases are different for each term, so they are relative phases and the state changes physically unless only one* E_n *appears.*

8.3 Time dependence of expectation values

Proposition 26 (Equation of motion)**.**

$$\frac{d}{dt}\langle A\rangle = \frac{i}{\hbar}\langle [H,A]\rangle + \langle \frac{\partial A}{\partial t}\rangle \, . \tag{69}$$

Proof. With $\langle A\rangle = \langle\psi(t)| \, A \, |\psi(t)\rangle$ and the chain rule,

$$\frac{d}{dt}\langle A\rangle = \left(\frac{d}{dt}\langle\psi(t)|\right) A \, |\psi(t)\rangle + \langle\psi(t)| \, A \left(\frac{d}{dt}|\psi(t)\rangle\right) + \langle\psi(t)| \, \frac{\partial A}{\partial t} \, |\psi(t)\rangle \, .$$

The Schrödinger equation and its bra (using $H^\dagger = H$) give $\frac{d}{dt}|\psi\rangle = -\frac{i}{\hbar}H|\psi\rangle$ and $\frac{d}{dt}\langle\psi| = \frac{i}{\hbar}\langle\psi|H$, so the first two terms are $\frac{i}{\hbar}\langle\psi|\,(HA - AH)\,|\psi\rangle$. □

Remark 44. *The* $\partial A/\partial t$ *term is only there if the operator itself has explicit time dependence, e.g. a time-dependent field or external force (Section 8.5).*

Definition 52 (Constant of the motion)**.** *An observable* A *with* $\partial A/\partial t = 0$ *and* $[H,A] = 0$*. Then* $\frac{d}{dt}\langle A\rangle = 0$*: it is a conserved quantity.*

Example. $A = H$ *always commutes with itself, so energy is conserved whenever* $\partial H/\partial t = 0$*.*

8.4 Spin precession

Put an electron ($q = -|e|$) in a uniform field $\vec{B} = B_0\hat{z}$. With $\vec{\mu} = -g\mu_B\vec{S}/\hbar$,

$$H_0 = -\vec{\mu}\cdot\vec{B} = \omega_0 S_z, \qquad \omega_0 = \frac{g\mu_B B_0}{\hbar}, \qquad [\omega_0] = \frac{[\text{energy}]}{[\hbar]} = [\text{frequency}]. \tag{70}$$

Since $[H_0, S_z] = 0$, S_z is a constant of the motion. The energy eigenstates are

$$H_0\,|\pm\hat{z}\rangle = \pm\frac{\hbar\omega_0}{2}\,|\pm\hat{z}\rangle \, .$$

Result 23 (Time evolution is a rotation)**.**

$$U(t) = e^{-i\omega_0 t S_z/\hbar} = R(\phi\hat{z})\Big|_{\phi=\omega_0 t}, \qquad U(t)\,|\pm\hat{z}\rangle = e^{\mp i\omega_0 t/2}\,|\pm\hat{z}\rangle \, . \tag{71}$$

Evolving under $H_0 = \omega_0 S_z$ *is the same as rotating about* $\hat{z}$ *by* $\phi = \omega_0 t$*. The states* $|\pm\hat{z}\rangle$ *are stationary, equivalently invariant under rotations about* $\hat{z}$*.*

Example (Starting in $|+\hat{x}\rangle$)**.** *$|+\hat{x}\rangle$ is a superposition of energy eigenstates, so by Eq.* (68)

$$|\psi(t)\rangle = e^{-iH_0 t/\hbar}\frac{1}{\sqrt{2}}\left(|+\hat{z}\rangle + |-\hat{z}\rangle\right) = \frac{1}{\sqrt{2}}\left(e^{-i\omega_0 t/2}\,|+\hat{z}\rangle + e^{i\omega_0 t/2}\,|-\hat{z}\rangle\right).$$

With $\langle +\hat{x}| \pm \hat{z}\rangle = \frac{1}{\sqrt{2}}$,

$$P_{+\hat{x}}(t) = |\langle +\hat{x}|\psi(t)\rangle|^2 = \left|\frac{1}{2}e^{-i\omega_0 t/2} + \frac{1}{2}e^{i\omega_0 t/2}\right|^2 = \cos^2\frac{\omega_0 t}{2} = \frac{1}{2}\left(1+\cos\omega_0 t\right). \tag{72}$$

8.4.1 Expectation values

With $S_x \xrightarrow{S_z} \frac{\hbar}{2}\sigma_x$ and $\psi(t) \xrightarrow{S_z} \frac{1}{\sqrt{2}}\left(e^{-i\omega_0 t/2}, e^{i\omega_0 t/2}\right)^T$,

$$\langle S_x(t)\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} e^{i\omega_0 t/2} & e^{-i\omega_0 t/2}\end{pmatrix}\frac{\hbar}{2}\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}\frac{1}{\sqrt{2}}\begin{pmatrix}e^{-i\omega_0 t/2}\\ e^{i\omega_0 t/2}\end{pmatrix}$$

$$= \frac{\hbar}{4}\left(e^{i\omega_0 t} + e^{-i\omega_0 t}\right) = \frac{\hbar}{2}\cos\omega_0 t.$$

Alternatively, from the probabilities,

$$\langle S_x(t)\rangle = \frac{\hbar}{2}\left(P_{+\hat{x}} - P_{-\hat{x}}\right) = \frac{\hbar}{2}\left(2P_{+\hat{x}} - 1\right) = \frac{\hbar}{2}\cos\omega_0 t.$$

The same matrix product with $S_y \xrightarrow{S_z} \frac{\hbar}{2}\sigma_y$ gives $\langle S_y(t)\rangle = \frac{\hbar}{2}\sin\omega_0 t$.

Result 24 (Precession)**.**

$$\langle \vec{S}(t)\rangle = \frac{\hbar}{2}\left(\cos\omega_0 t\,\hat{x} + \sin\omega_0 t\,\hat{y}\right), \qquad \langle S_z\rangle = 0. \tag{73}$$

The average spin rotates counterclockwise about $\hat{z}$ at angular frequency ω_0 (Larmor precession), while $\langle S_z\rangle$ stays constant as a constant of the motion should.

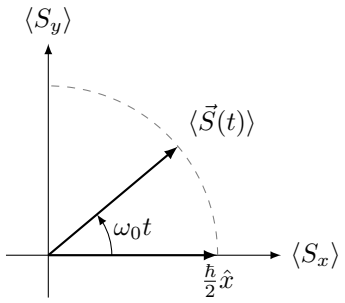


Figure 3: Precession of $\langle \vec{S}\rangle$ in the x-y plane, starting from $|+\hat{x}\rangle$.

8.4.2 From the equation of motion

The same result follows without ever writing the state. Using $[S_z, S_x] = i\hbar S_y$ and $[S_z, S_y] = -i\hbar S_x$,

$$\frac{d}{dt}\langle S_x\rangle = \frac{i}{\hbar}\langle[\omega_0 S_z, S_x]\rangle = -\omega_0\langle S_y\rangle, \tag{74}$$

$$\frac{d}{dt}\langle S_y\rangle = \frac{i}{\hbar}\langle[\omega_0 S_z, S_y]\rangle = \omega_0\langle S_x\rangle. \tag{75}$$

Differentiating Eq. (74) and using Eq. (75),

$$\frac{d^2}{dt^2}\langle S_x\rangle = -\omega_0\frac{d}{dt}\langle S_y\rangle = -\omega_0^2\langle S_x\rangle,$$

with general solution $\langle S_x\rangle = A\cos\omega_0 t + B\sin\omega_0 t$. The initial conditions are $\langle S_x(0)\rangle = \hbar/2$ and, from Eq. (74), $\frac{d}{dt}\langle S_x\rangle\big|_{t=0} = -\omega_0\langle S_y(0)\rangle = 0$, so $A = \hbar/2$, $B = 0$. Then $\langle S_y\rangle = -\frac{1}{\omega_0}\frac{d}{dt}\langle S_x\rangle$ gives

$$\langle S_x(t)\rangle = \frac{\hbar}{2}\cos\omega_0 t, \qquad \langle S_y(t)\rangle = \frac{\hbar}{2}\sin\omega_0 t.$$

8.5 Magnetic resonance

Definition 53 (Magnetic resonance)**.** *Driving transitions between spin states with an oscillating field tuned to the precession frequency. Applications:*

- Nuclear magnetic resonance *(NMR), on nuclear spin $\vec{I}$: e.g. medical imaging (MRI), which maps local water content through resonant absorption of radio-frequency (∼MHz) radiation.*
- Electron spin resonance *(ESR), on electron spin $\vec{S}$: e.g. characterizing materials with microwave (∼GHz) radiation.*
- Qubit control*: single-qubit gates (unitary operations) on qubits made from electron spins, nuclear spins, ions, neutral atoms or superconducting circuits.*

The field is a strong longitudinal part plus a weaker oscillating transverse part,

$$\vec{B}(t) = B_0\hat{z} + B_1\cos\omega t\,\hat{x}, \qquad B_0 \gg B_1, \tag{76}$$

so the Hamiltonian now has explicit time dependence:

$$H(t) = -\vec{\mu}\cdot\vec{B}(t) = \omega_0 S_z + \Omega\cos(\omega t)\,S_x, \tag{77}$$

with, using the magnetons defined with the Stern-Gerlach experiment,

$$\text{ESR } (q = -|e|): \quad \omega_0 = \frac{g\mu_B B_0}{\hbar}, \qquad \Omega = \frac{g\mu_B B_1}{\hbar},$$

$$\text{NMR } (q = +|e|): \quad \omega_0 = -\frac{g_I\mu_N B_0}{\hbar}, \qquad \Omega = -\frac{g_I\mu_N B_1}{\hbar}.$$

For $B_0 \sim 1$ T, $\omega_0/2\pi$ is in the GHz range for ESR ($g \approx 2$) and the MHz range for NMR, since $\mu_N \approx \mu_B/1836$.

Remark 45 (Angular vs linear frequency)**.** *ω_0 is an angular frequency, in rad/s. $\omega_0/2\pi$ is the linear frequency, in Hz = cycles/s, with 1 cycle = 2π rad.*

Remark 46. *With H time-dependent, $U(t) = e^{-iHt/\hbar}$ no longer holds, and we must work with the Schrödinger equation directly, $\partial_t |\psi(t)\rangle = -\frac{i}{\hbar} H(t) |\psi(t)\rangle$.*

8.5.1 The rotating frame

Split the Hamiltonian into a large static part and a small driving part,

$$H(t) = H_0 + V(t), \qquad H_0 = \omega_0 S_z, \qquad V(t) = \Omega \cos(\omega t)\, S_x.$$

Since $\omega_0 \gg \Omega$, most of the evolution is the fast precession from H_0. Factor it out:

$$|\psi(t)\rangle = U_0(t)\, |\tilde{\psi}(t)\rangle\,, \qquad U_0(t) = e^{-iH_0 t/\hbar}, \tag{78}$$

where U_0 is the "fast" part and $|\tilde{\psi}(t)\rangle$ the "slow" part.

Proposition 27 (Schrödinger equation in the rotating frame)**.**

$$\partial_t |\tilde{\psi}(t)\rangle = -\frac{i}{\hbar} \tilde{V}(t)\, |\tilde{\psi}(t)\rangle\,, \qquad \tilde{V}(t) = U_0^\dagger(t) V(t) U_0(t). \tag{79}$$

Proof. Substituting into the Schrödinger equation with $\dot{U}_0 = -\frac{i}{\hbar} H_0 U_0$,

$$-\frac{i}{\hbar} H_0 U_0\, |\tilde{\psi}\rangle + U_0\, \partial_t |\tilde{\psi}\rangle = -\frac{i}{\hbar} (H_0 + V)\, U_0\, |\tilde{\psi}\rangle\,.$$

The H_0 terms cancel. Multiplying on the left by $U_0^\dagger$ gives the result. □

Definition 54 (Rotating frame)**.** *Since $U_0(t) = e^{-i\omega_0 t S_z/\hbar} = R(\omega_0 t\, \hat{z})$ is a rotation about $\hat{z}$ by the time-dependent angle $\omega_0 t$, describing the system by $|\tilde{\psi}(t)\rangle$ is called going into the rotating frame.*

To compute $\tilde{V}$, write $S_x = \frac{1}{2}(S_+ + S_-)$ with the spin-$\frac{1}{2}$ ladder operators (Section 7.5),

$$S_+ = \hbar\, |+\hat{z}\rangle \langle -\hat{z}|\,, \qquad S_- = \hbar\, |-\hat{z}\rangle \langle +\hat{z}|\,.$$

Lemma 4.

$$e^{i\omega_0 t S_z/\hbar}\, S_\pm\, e^{-i\omega_0 t S_z/\hbar} = e^{\pm i\omega_0 t} S_\pm. \tag{80}$$

Proof. For S_+: $e^{-i\omega_0 t S_z/\hbar}$ acting right on $|-\hat{z}\rangle$ gives $e^{i\omega_0 t/2}$, and $e^{i\omega_0 t S_z/\hbar}$ acting left on $|+\hat{z}\rangle$ gives $e^{i\omega_0 t/2}$, a total of $e^{i\omega_0 t}$. Swapping the roles gives $e^{-i\omega_0 t}$ for S_-. □

With $\cos\omega t = \frac{1}{2}\left(e^{i\omega t} + e^{-i\omega t}\right)$,

$$\begin{aligned} \tilde{V}(t) &= \frac{\Omega}{4} \left(e^{i\omega t} + e^{-i\omega t}\right) \left(e^{i\omega_0 t} S_+ + e^{-i\omega_0 t} S_-\right) \\ &= \frac{\Omega}{4} \Big(\underbrace{e^{-i(\omega-\omega_0)t} S_+ + e^{i(\omega-\omega_0)t} S_-}_{\text{slow for } \omega \approx \omega_0} + \underbrace{e^{i(\omega+\omega_0)t} S_+ + e^{-i(\omega+\omega_0)t} S_-}_{\text{fast}} \Big). \end{aligned} \tag{81}$$

8.5.2 Rotating-wave approximation

Integrating the rotating-frame equation,

$$|\tilde{\psi}(t)\rangle - |\tilde{\psi}(0)\rangle = -\frac{i}{\hbar}\int_0^t dt'\,\tilde{V}(t')\,|\tilde{\psi}(t')\rangle\,.$$

The fast terms $\sim e^{\pm i(\omega+\omega_0)t}$ oscillate many times while $|\tilde{\psi}\rangle$ barely changes, so they average to zero on a time scale $t \gtrsim 1/(\omega+\omega_0)$. Near resonance ($\omega \approx \omega_0$) the slow terms $\sim e^{\pm i(\omega-\omega_0)t}$ dominate.

Definition 55 (Rotating-wave approximation (RWA))**.** *Dropping the terms* $\sim e^{\pm i(\omega+\omega_0)t}$*. The corrections are small when* $\Omega/\omega_0 \ll 1$*, i.e.* $B_0 \gg B_1$*. Then*

$$\tilde{V}(t) \simeq \frac{\Omega}{4}\left(e^{-i\delta t}S_+ + e^{i\delta t}S_-\right). \tag{82}$$

Definition 56 (Detuning)**.** $\delta = \omega - \omega_0$*, how far the drive is from resonance.*

Result 25 (On resonance)**.** *For* $\omega = \omega_0$*,* $\delta = 0$ *and* $\tilde{V}$ *is time-independent:*

$$\tilde{V} \simeq \frac{\Omega}{4}(S_+ + S_-) = \frac{\Omega}{2}S_x, \qquad |\tilde{\psi}(t)\rangle \simeq e^{-i\Omega t S_x/2\hbar}\,|\psi(0)\rangle = R\left(\tfrac{\Omega t}{2}\hat{x}\right)|\psi(0)\rangle\,, \tag{83}$$

using $|\tilde{\psi}(0)\rangle = |\psi(0)\rangle$ *since* $U_0(0) = \mathbb{1}$*. In the rotating frame the spin simply rotates about* $\hat{x}$ *at angular frequency* $\Omega/2$*.*

8.5.3 Rabi oscillations

Start in $|\psi(0)\rangle = |+\hat{z}\rangle = \frac{1}{\sqrt{2}}(|+\hat{x}\rangle + |-\hat{x}\rangle)$, i.e. expand in the eigenstates of $\tilde{V} \propto S_x$ as in Eq. (68). With $S_x\,|\pm\hat{x}\rangle = \pm\frac{\hbar}{2}\,|\pm\hat{x}\rangle$,

$$|\tilde{\psi}(t)\rangle \simeq \frac{1}{\sqrt{2}}\left(e^{-i\Omega t/4}\,|+\hat{x}\rangle + e^{i\Omega t/4}\,|-\hat{x}\rangle\right)$$

$$|\tilde{\psi}(t)\rangle \xrightarrow{S_x} \frac{1}{\sqrt{2}}\begin{pmatrix} e^{-i\Omega t/4} \\ e^{i\Omega t/4}\end{pmatrix}.$$

Since U_0 is a function of S_z, $[U_0, S_z] = 0$, so

$$\langle S_z\rangle = \langle\psi(t)|\,S_z\,|\psi(t)\rangle = \langle\tilde{\psi}(t)|\,S_z\,|\tilde{\psi}(t)\rangle\,.$$

With $S_z \xrightarrow{S_x} \frac{\hbar}{2}\sigma_x$ (problem "$\hat{S}_z$ in the S_x basis"),

$$\langle S_z\rangle = \frac{1}{2}\begin{pmatrix} e^{i\Omega t/4} & e^{-i\Omega t/4}\end{pmatrix}\frac{\hbar}{2}\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}\begin{pmatrix} e^{-i\Omega t/4} \\ e^{i\Omega t/4}\end{pmatrix} = \frac{\hbar}{4}\left(e^{i\Omega t/2} + e^{-i\Omega t/2}\right).$$

Result 26 (Rabi oscillations)**.**

$$\langle S_z(t)\rangle = \frac{\hbar}{2}\cos\frac{\Omega t}{2}, \qquad P_{+\hat{z}}(t) = \cos^2\frac{\Omega t}{4}. \tag{84}$$

The spin flips back and forth between $|+\hat{z}\rangle$ *and* $|-\hat{z}\rangle$*. A drive left on for* $\Omega t/2 = \pi$ *(a "π-pulse") takes* $|+\hat{z}\rangle$ *to* $|-\hat{z}\rangle$*: this is how a spin qubit is flipped.*

Remark 47 (WARNING: factor of 2)**.** *Bill's notes write the phases as* $e^{\mp i\Omega t/2}$ *and get* $\langle S_z\rangle = \frac{\hbar}{2}\cos\Omega t$*. The eigenvalues of* S_x *are* $\pm\hbar/2$*, not* $\pm\hbar$*, so the phases are* $e^{\mp i\Omega t/4}$ *and the Rabi frequency is* $\Omega/2$*. A numerical solution of the full* $H(t)$ *confirms* $\cos(\Omega t/2)$*. The half comes from the RWA:* $\cos\omega t$ *splits into two counter-rotating pieces and only one of them is kept.*

A Solutions

Solution 1 (The exponential as a limit)**.** *Take the logarithm:*

$$\ln\left(1+\frac{x}{N}\right)^N = N\ln\left(1+\frac{x}{N}\right).$$

With $\ln(1+u) = u - \frac{u^2}{2} + \frac{u^3}{3} - \cdots$ *and* $u = x/N$*,*

$$N\ln\left(1+\frac{x}{N}\right) = x - \frac{x^2}{2N} + \frac{x^3}{3N^2} - \cdots.$$

Every term after the first vanishes as $N\to\infty$*, so the logarithm tends to* x *and*

$$\lim_{N\to\infty}\left(1+\frac{x}{N}\right)^N = e^x.$$

Solution 2 (Other bases)**.**

$$\begin{aligned}
|+\hat{z}\rangle &\xrightarrow{S_z} \begin{pmatrix}1\\0\end{pmatrix},\\
|+\hat{z}\rangle &\xrightarrow{S_x} \begin{pmatrix}\langle+\hat{x}|+\hat{z}\rangle\\ \langle-\hat{x}|+\hat{z}\rangle\end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix},\\
|+\hat{z}\rangle &\xrightarrow{S_y} \begin{pmatrix}\langle+\hat{y}|+\hat{z}\rangle\\ \langle-\hat{y}|+\hat{z}\rangle\end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix}.
\end{aligned}$$

Solution 3 (A probability by matrix mechanics)**.**

$$\psi \xrightarrow{S_z} \frac{1}{\sqrt{5}}\begin{pmatrix}1\\2i\end{pmatrix}, \qquad \langle+\hat{x}| \xrightarrow{S_z} \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\end{pmatrix},$$

$$\langle+\hat{x}|\psi\rangle = \frac{1+2i}{\sqrt{10}}, \qquad P\big(S_x = +\tfrac{\hbar}{2}\big) = \frac{|1+2i|^2}{10} = \frac{1}{2}.$$

Take the modulus of the whole sum, not of each term separately.

Solution 4 (The matrix of $\hat{S}_z$)**.** *Every overlap is* 0 *or* 1*, so only the diagonal survives:*

$$\langle\pm\hat{z}|\,\hat{S}_z\,|\pm\hat{z}\rangle = \pm\frac{\hbar}{2}, \qquad \hat{S}_z \xrightarrow{S_z} \frac{\hbar}{2}\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}.$$

An operator is diagonal in its own eigenbasis, with its outcomes on the diagonal.

Solution 5 (The matrix of $\hat{S}_x$)**.** *The overlaps are* $\langle+\hat{z}|\pm\hat{x}\rangle = \langle-\hat{z}|+\hat{x}\rangle = \frac{1}{\sqrt{2}}$ *and* $\langle-\hat{z}|-\hat{x}\rangle = -\frac{1}{\sqrt{2}}$*, all real. Then*

$$\begin{aligned}
\langle+\hat{z}|\,\hat{S}_x\,|+\hat{z}\rangle &= \frac{\hbar}{2}\left(\tfrac{1}{2}-\tfrac{1}{2}\right) = 0, & \langle+\hat{z}|\,\hat{S}_x\,|-\hat{z}\rangle &= \frac{\hbar}{2}\left(\tfrac{1}{2}+\tfrac{1}{2}\right) = \frac{\hbar}{2},\\
\langle-\hat{z}|\,\hat{S}_x\,|+\hat{z}\rangle &= \frac{\hbar}{2}\left(\tfrac{1}{2}+\tfrac{1}{2}\right) = \frac{\hbar}{2}, & \langle-\hat{z}|\,\hat{S}_x\,|-\hat{z}\rangle &= \frac{\hbar}{2}\left(\tfrac{1}{2}-\tfrac{1}{2}\right) = 0,
\end{aligned}$$

where the + signs come from the minus in $\langle -\hat{z}| - \hat{x}\rangle$ meeting the minus in front of the second projector. So

$$\hat{S}_x \xrightarrow{S_z} \frac{\hbar}{2}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Solution 6 (The matrix of $\hat{S}_y$)**.** *The overlaps are complex and the bra side is conjugated: $\langle -\hat{z}| \pm \hat{y}\rangle = \pm\frac{i}{\sqrt{2}}$, $\langle \pm\hat{y}| - \hat{z}\rangle = \mp\frac{i}{\sqrt{2}}$, and $\langle +\hat{z}| \pm \hat{y}\rangle = \langle \pm\hat{y}| + \hat{z}\rangle = \frac{1}{\sqrt{2}}$. Then*

$$\langle +\hat{z}|\,\hat{S}_y\,|-\hat{z}\rangle = \frac{\hbar}{2}\left(\tfrac{1}{\sqrt{2}}\cdot\tfrac{-i}{\sqrt{2}} - \tfrac{1}{\sqrt{2}}\cdot\tfrac{i}{\sqrt{2}}\right) = -\frac{i\hbar}{2},$$
$$\langle -\hat{z}|\,\hat{S}_y\,|+\hat{z}\rangle = \frac{\hbar}{2}\left(\tfrac{i}{\sqrt{2}}\cdot\tfrac{1}{\sqrt{2}} - \tfrac{-i}{\sqrt{2}}\cdot\tfrac{1}{\sqrt{2}}\right) = +\frac{i\hbar}{2},$$

and the diagonal entries vanish because $i\cdot(-i) = 1$ in both terms. So

$$\hat{S}_y \xrightarrow{S_z} \frac{\hbar}{2}\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}.$$

Solution 7 (Expectation value and uncertainty by matrix mechanics)**.** *In the S_z basis,*

$$\psi \xrightarrow{S_z} \frac{1}{\sqrt{10}}\left[\begin{pmatrix}1\\ i\end{pmatrix} + \begin{pmatrix}2i\\ 2\end{pmatrix}\right] = \frac{1}{\sqrt{10}}\begin{pmatrix}1+2i\\ 2+i\end{pmatrix}.$$

Then

$$\langle S_z\rangle = \psi^\dagger \hat{S}_z \psi = \frac{\hbar}{2}\cdot\frac{1}{10}\begin{pmatrix}1-2i & 2-i\end{pmatrix}\begin{pmatrix}1+2i\\ -(2+i)\end{pmatrix} = \frac{\hbar}{2}\cdot\frac{5-5}{10} = 0.$$

Since $\hat{S}_z^2 = \frac{\hbar^2}{4}\mathbb{1}$, $\langle S_z^2\rangle = \frac{\hbar^2}{4}$, so $\Delta S_z = \hbar/2$: both outcomes have probability $\frac{1}{2}$ and the spread is maximal.

Solution 8 (Expectation value from the S_x basis)**.**

$$|\psi\rangle \xrightarrow{S_x} \psi^x = \frac{1}{\sqrt{3}}\begin{pmatrix}1\\ i\sqrt{2}\end{pmatrix}.$$

Change to the S_z basis with U from Eq. (37):

$$\psi^z = U\psi^x = \frac{1}{\sqrt{6}}\begin{pmatrix}1+i\sqrt{2}\\ 1-i\sqrt{2}\end{pmatrix}, \qquad \psi^{z\dagger} = \frac{1}{\sqrt{6}}\begin{pmatrix}1-i\sqrt{2} & 1+i\sqrt{2}\end{pmatrix}.$$

Then

$$\langle S_z\rangle = \psi^{z\dagger}\hat{S}_z\psi^z = \frac{\hbar}{2}\cdot\frac{1}{6}\left(|1+i\sqrt{2}|^2 - |1-i\sqrt{2}|^2\right) = \frac{\hbar}{2}\left(\frac{3}{6} - \frac{3}{6}\right) = 0.$$

Both components have equal modulus, so both S_z outcomes have probability $\frac{1}{2}$.

Solution 9 ($\hat{S}_z$ in the S_x basis)**.** *U from Eq. (37) is real and symmetric, so $U^\dagger = U$ and*

$$\hat{S}_z \xrightarrow{S_x} U^\dagger D\, U = \frac{\hbar}{2}\cdot\frac{1}{2}\begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix}\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}\begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix} = \frac{\hbar}{2}\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}.$$

This is $\frac{\hbar}{2}\sigma_x$: in the S_x basis, $\hat{S}_z$ looks the way $\hat{S}_x$ looks in the S_z basis.

Keeping the state in the S_x basis,

$$\langle S_z\rangle = \psi^{x\dagger}\hat{S}_z^x\psi^x = \frac{\hbar}{2}\cdot\frac{1}{3}\begin{pmatrix}1 & -i\sqrt{2}\end{pmatrix}\begin{pmatrix}i\sqrt{2}\\ 1\end{pmatrix} = \frac{\hbar}{2}\cdot\frac{i\sqrt{2}-i\sqrt{2}}{3} = 0,$$

matching the previous solution. In general, for $\psi^x = (c_+, c_-)^T$ this gives $\langle S_z\rangle = \hbar\,\mathrm{Re}\left(c_+^* c_-\right)$, which vanishes here because the relative phase is i.

Solution 10 ($R(\phi\hat{z})$ in the S_x basis)**.** *Expanding the S_x states in $\hat{z}$,*

$$R\,|\pm\hat{x}\rangle = \frac{1}{\sqrt{2}}\left(e^{-i\phi/2}\,|+\hat{z}\rangle \pm e^{i\phi/2}\,|-\hat{z}\rangle\right).$$

Hitting these with $\langle\pm\hat{x}| = \frac{1}{\sqrt{2}}\left(\langle+\hat{z}| \pm \langle-\hat{z}|\right)$ and using Euler's formula,

$$\langle+\hat{x}|\,R\,|+\hat{x}\rangle = \langle-\hat{x}|\,R\,|-\hat{x}\rangle = \frac{1}{2}\left(e^{-i\phi/2}+e^{i\phi/2}\right) = \cos\frac{\phi}{2},$$
$$\langle+\hat{x}|\,R\,|-\hat{x}\rangle = \langle-\hat{x}|\,R\,|+\hat{x}\rangle = \frac{1}{2}\left(e^{-i\phi/2}-e^{i\phi/2}\right) = -i\sin\frac{\phi}{2},$$

so

$$R(\phi\hat{z}) \xrightarrow{S_x} \begin{pmatrix}\cos\frac{\phi}{2} & -i\sin\frac{\phi}{2}\\ -i\sin\frac{\phi}{2} & \cos\frac{\phi}{2}\end{pmatrix}.$$

Check: at $\phi = \pi/2$, $R\,|+\hat{x}\rangle \xrightarrow{S_x} \frac{1}{\sqrt{2}}(1,-i)^T$, which is $|+\hat{y}\rangle$ up to a global phase, as rotating $\hat{x}$ by $90°$ about $\hat{z}$ should give.

B Useful Links

Stern-Gerlach experiment:
https://www.youtube.com/shorts/QG7kJXxN1Kk